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Synfolding Magnetization in the Jurassic Preuss Sandstone, Wyoming-Idaho-Utah Thrust Belt

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The Jurassic Preuss Sandstone, exposed in five thrust plates of the Wyoming-Idaho-Utah thrust belt, carries directions of remanent magnetization that group most tightly after only partial unfolding. Field, petrographic, and rock magnetic evidence indicates that the carrier of this magnetization is detrital, low-Ti titanomagnetite. The detrital titanomagnetite was remagnetized at low temperatures (75°–150°C) probably completely during folding. Anisotropy of magnetic susceptibility and petrographic observations indicate that the detrital titanomagnetite has been affected by tectonic strain. We suggest that low-temperature remagnetization of the detrital titanomagnetite was either a viscous partial thermoremanent magnetization, the acquisition of which was enhanced by stress, or a piezoremanent magnetization that involved stress-induced movement of domain walls during intracrystalline strain, or was a combination of the two mechanisms. Stress may promote remagnetization at temperatures much lower than predicted by current theoretical models. Other mechanisms, such as acquisition of chemical remanent magnetization during folding, deflection of a prefolding magnetization by internal strain, or combination of components of magnetization with different direction cannot account for the geometry of magnetization in the Preuss. The locus of acquisition of synfolding magnetization in the Preuss migrated in conjunction with deformation in the thrust belt. A model is presented in which synfolding magnetization was acquired during cooling and folding as strata moved up thrust ramps. A lack of reverse-polarity directions remains a puzzling feature of the remanence. The remanent direction is tentatively interpreted to reflect the predominant polarity state during its acquisition over an extended rather than a discrete time period during folding in Late Cretaceous and early Tertiary (?) periods of predominantly normal polarity.

INTRODUCTION

Several recent paleomagnetic studies from orogenic belts on the North American and Eurasian continents show directions of remanent magnetization in folded rocks that group most closely after only partial correction for structural tilt [*Russell et al.*, 1982; *McClelland-Brown*, 1983; *McCabe et al.*, 1983; *Schwartz and Van der Voo*, 1984; *Kent and Opdyke*, 1985; *Kent*, 1988; *Miller and Kent*, 1986*a*, *b*, 1988*a*, 1989; *Granier et al.*, 1986; *Van Alstine*, 1986; *Benthien and Elmore*, 1987; *Cogne' and Perroud*, 1987; *Gales et al.*, 1987; *Vick et al.*, 1987; *Bachtadse et al.*, 1987]. This behavior, or geometric trait, could be produced if the magnetization was acquired during folding (i.e., it is synfolding magnetization), although it could be produced by other mechanisms as well. The origin of this type of magnetization is not well understood. Current models for its genesis include the acquisition of a secondary magnetization during folding, such as chemical remanent magnetization (CRM) [e.g., *McCabe et al.*, 1983; *Van Alstine*, 1986], thermochemical remanent magnetization (TCRM) [*Granier et al.*, 1986], or the structural modification of a prefolding magnetization [*van der Pluijm*, 1986, 1987; *Cogne' and Perroud*, 1987; *Kodama*, 1988]. Remanent magnetization whose directions group most closely at partial untilting can yield unique information on the chemical, thermal, or structural evolution of the host rocks, but the origin of this type of magnetization must be better understood before such information can be interpreted. In this paper we examine the origin of remanent magnetization in the Jurassic Preuss Sandstone, Wyoming-Idaho-Utah thrust belt, which we believe was acquired during folding. Constraints on the chemical, thermal, and structural history of the Preuss are integrated with its magnetic properties to enable a close evaluation of the origin of this magnetization. The proposed origin of synfolding magnetization in the Preuss may be applicable to synfolding magnetization in other rock units.

The Preuss is a source of aeromagnetic anomalies in the west central portions of the thrust belt [*Fishman et al.*, 1989]. We became interested in the Preuss because it has been proposed that hydrocarbon seepage might promote growth of diagenetic magnetite and thus cause magnetic anomalies in sedimentary rock over potential hydrocarbon accumulations [*Donovan et al.*, 1979]. Because aeromagnetic anomalies associated with the Preuss are on trend with hydrocarbon accumulations buried beneath younger Tertiary strata in the southern portions of the thrust belt, we examined the Preuss to determine if it contained diagenetic magnetite that might produce the anomalies. It has also been suggested that hydrocarbon-induced growth of diagnetic magnetite might cause CRM-related synfolding magnetization in some rocks [*McCabe et al.*, 1983; *Oliver*, 1986; *Benthien and Elmore*, 1987]. Recognition of synfolding magnetization in the Preuss heightened our curiosity about the nature of the magnetization in the Preuss. More information on the relevance of this study to the proposed association among aeromagnetic anomalies, diagenetic magnetite, and hydrocarbon seepage can be found in the work of *Fishman et al.* [1989].

In this paper we show that synfolding magnetization in the Preuss is carried by detrital low-Ti titanomagnetite grains that were reset at temperatures (<150°C) much lower than predicted by theoretical models, based on the work of *Pullaiah et al.* [1975] or of *Walton* [1982]. Neither the synfolding remanent magnetization nor the aeromagnetic

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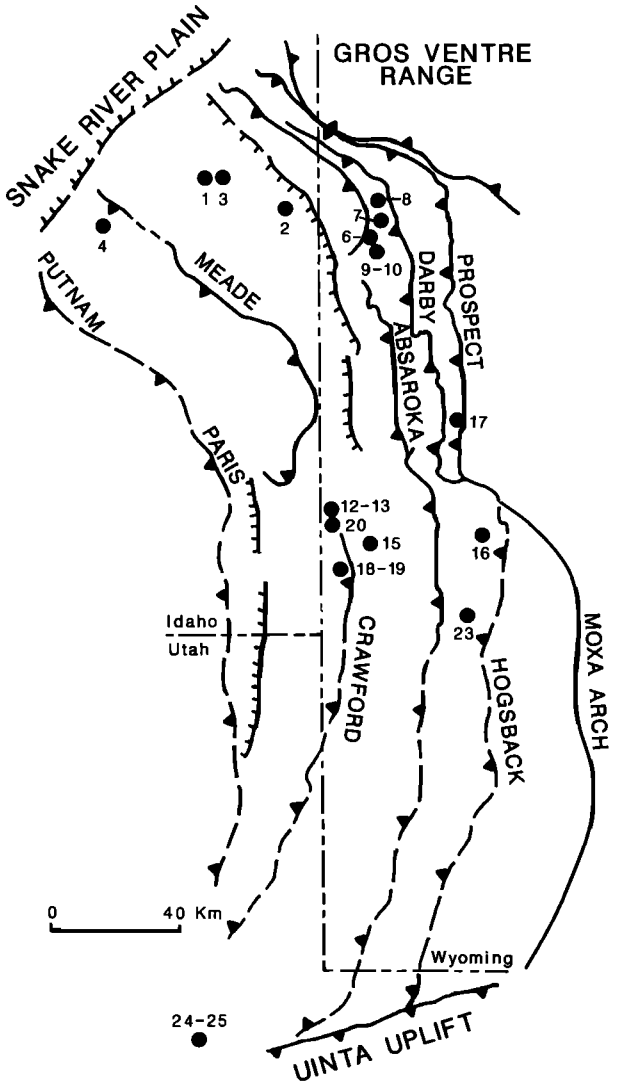


Fig. 1. Generalized tectonic map of Wyoming-Idaho-Utah thrust belt showing major thrust plates, normal faults, and locations of sites (Table 1) studied in the paleomagnetic analysis of the Preuss Sandstone.

anomalies associated with the Preuss Sandstone are related to growth of diagenetic magnetite. No evidence for the presence of diagenetic magnetite in the Preuss was found. Finally, we suggest that remagnetization of the detrital titanomagnetite in the Preuss was produced by either a stress-enhanced acquisition of a viscous partial thermoremanent magnetization (VpTRM) or by stress-induced movement of domain walls resulting from intracrystalline strain within the magnetite grains (piezoremanent magnetization, PRM) or by a combination of the two processes.

Geologic Setting

The thrust belt of western Wyoming, eastern Idaho, and northeastern Utah (Figure 1) is composed of numerous thrust plates that imbricate Phanerozoic sedimentary strata of the western North America miogeosyncline. Clastic strata were shed from the rising thrust plates into a subsiding foreland basin and then incorporated into subsequent thrust plates. The strata record a general progression of thrusting from west to east across the belt [*Armstrong and Oriel*, 1965; *Wiltschko and Dorr*, 1983]. Major thrusts typically formed on the foreland side of preexisting thrusts, carrying the preexisting thrusts in their hanging wall in a piggyback fashion [*Royse et al.*, 1975; *Woodward*, 1986]. Thrusting probably began with movement on the westernmost Paris-Putnam thrust system in late Early Cretaceous [*Heller et al.*, 1986], although perhaps as early as Late Jurassic [*Armstrong and Oriel*, 1965; *Wiltschko and Dorr*, 1983]. Successive movements occurred on the Meade-Crawford (the Meade thrust may be slightly older than the Crawford thrust [*Evans and Craddock*, 1985]) and Absaroka thrusts in Late Cretaceous. Timing and correlations among eastern thrusts (Darby, Hogsback, Prospect) are more complicated. Movement on the Hogsback thrust during Paleocene time and on the Prospect thrust near Hoback Basin during the early Eocene time has been documented [*Wiltschko and Dorr*, 1983]. *Kraig et al.* [1987] have proposed, however, an ancestral phase of movement along the Prospect thrust in response to movement of the Moxa Arch in Late Cretaceous. The relative timing of movement on the Darby and Prospect thrusts and a connection between the Darby and Hogsback thrusts are debated [*Royse et al.*, 1975; *Dixon*, 1982; *Wiltschko and Dorr*, 1983; *Woodward*, 1986; *Kraig et al.*, 1987]. The durations of movement on individual thrusts are constrained to different levels of confidence [*Wiltschko and Dorr*, 1983]. The Prospect thrust at Hoback Basin and the Darby thrust (if part of the Hogsback thrust) are constrained to have moved for less than about 2 and 6 m.y., respectively. The Absaroka thrust, which has greater displacement, was probably active over a longer time period (about 17 m.y.), but periodic deposition of synorogenic conglomerates derived from the thrust plate suggests that the thrust plate moved during several separate pulses. The duration of major movement on the Crawford-Meade thrust system may be less than 10 m.y. if its movement had slowed by the time of initial movement on the underlying Absaroka thrust. Shortening within the major thrust plates was taken up by folds and small-displacement thrusts. Most major folds are predominantly fault propagation or fault bend folds that are genetically related to thrust movement [e.g., *Royse et al.*, 1975; *Boyer and Elliot*, 1982; *Boyer*, 1986]. Therefore, in general, the folds should be younger in the east than in the west, as are the thrust plates in which they occur, although this relation may not hold within an individual thrust plate. Timing relations among folds and thrusts may be more complicated in the eastern plates.

The Callovian Preuss Sandstone is predominantly a clastic unit of mudstone, siltstone, and sandstone in the northern portions of the thrust belt [*Imlay*, 1952; *Hileman*, 1973]. Evaporite deposits comprise the lower portions of the Preuss in the southwestern portions of the thrust belt [*Mansfield*, 1927]. The formation thins from west to east [*Jordan*, 1985] and is thought to have been deposited in near-shore marine, tidal, or sabkha depositional environments [*Hileman*, 1973; *Jordan*, 1985]. For this study we sampled widely in the thrust belt (Figure 1) to test for possible geographic variations in its magnetic properties. From west to east, sites are located in the Meade, Crawford, Absaroka, Darby, and Hogsback thrust plates. Our sampling was concentrated in the central and northern portions of the thrust belt because much of the southern portion of the thrust belt is covered by Tertiary strata, although samples were collected at two

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southern sites (24, 25) from Emigration Canyon near Salt Lake City, Utah.

Constraints on the thermal history of the Preuss are particularly important for the interpretations in this study. Values of vitrinite reflectance of plant material separated from the Preuss Sandstone at seven sites range from 0.54 to 0.99 [*Fishman et al.*, 1989]. These values indicate that the Preuss was subjected to maximum temperatures of 75°–150°C, based on the temperature-vitrinite reflectance correlation of *Barker and Pawlewicz* [1986]. Values of conodont alteration indices (1.5–2) in underlying Triassic strata (A. E. Harris, written communication, 1985) are consistent with these temperatures. Before the thrust belt deformation the Preuss was overlain only by thin deposits of the Middle and Upper Jurassic Stump Formation (100 m maximum [*Jordan*, 1985]) and perhaps the Late Jurassic lower portions of the Ephraim Conglomerate (about 100 m [*Heller et al.*, 1986]). Assuming an ambient surface temperature of 25°C and a geothermal gradient of 30°C km^{-1}, it is unlikely that the Preuss experienced temperatures exceeding 35°C before initiation of thrusting. Instead, the Preuss probably experienced maximum temperatures during the deformation. Thermal models for overthrust belts [*Angevine and Turcotte*, 1983; *Furlong and Edman*, 1984] suggest that a point in the Preuss that was buried beneath an overriding thrust plate would reach its maximum temperature soon (0–10 m.y., depending on thrust velocity and thrust plate thickness) after thrusting ceased in response to thermal reequilibration of tectonically disturbed isotherms. A point farther to the east in the foreland basin, which would subside in response to thrust loading [*Jordan*, 1981], would be heated during burial beneath synorogenic sediments. In either setting, subsequent incorporation of the point into the next thrust plate would cause cooling at the point as it was uplifted toward the surface. Once uplifted, it is unlikely that the point would be buried and reheated again. Therefore it was probable that the Preuss at a given point achieved its maximum temperature and then cooled in less than 10 m.y. during a progressive deformation.

Field and Laboratory Methods

In the field, samples for the paleomagnetic study of the Preuss were either collected with a portable drill (290 samples) or as oriented blocks (51 samples). Samples were oriented with both magnetic and solar compasses, but no significant difference was found between magnetic and solar azimuths. At most sites only sandstone layers were sufficiently coherent for sampling, although a few beds of siltstone and shale were sampled at sites 9, 16, and 23. From 8 to 40 independent oriented samples were collected at each site. At sites where oriented blocks were collected, fewer independently oriented samples were taken, but several specimens were drilled from each block in the laboratory to provide sufficient material for demagnetization studies. The quality of exposure and number of sandstone layers at a site controlled the stratigraphic coverage of sampling at each site. Typically three to four sandstone layers within a 10–15-m stratigraphic interval were sampled at each site. Generally, more layers were sampled in the western sites (12, 13, 18, 19, 20) where the proportion of sandstone in the Preuss appears to be greater than in more eastern sites. Samples were collected on both limbs of individual folds in five areas within either the Crawford, Absaroka, or Darby thrust plates. Individual fold tests could only be carried out for four of the folds because one site (25) from the southernmost fold in Emigration Canyon yielded scattered magnetization directions. In total, 434 specimens from 341 samples collected from 20 sites provide the data for our study.

In the laboratory the remanent magnetization of the specimens were measured using a spinner magnetometer with a sensitivity of about 1×10^{-4} A m^{-1}. Demagnetization was performed using either a two-axis tumbling degausser capable of peak inductions of 100 mT or an oven shielded from the Earth's magnetic field by six nested mu-metal shields. Residual fields within the cooling chamber of the oven were <10 nT. All demagnetization was performed in a stepwise manner. Typically, the number of steps for each demagnetization sequence ranged from 8 to 17, and the number of steps was adjusted to fit the blocking temperature or coercivity spectra of the sample and the complexity of the magnetization. The results from the demagnetization sequence of each specimen were plotted on orthogonal vector plots, and a least squares fit of linear segments containing four or more points was used to determine the orientation of the characteristic component of magnetization in the specimen. Fisher statistics [*Fisher*, 1953] were used to compute the mean directions. Incremental correction of the data for the tectonic tilt at each site was performed as a series of 50 2% untilting steps. Free demagnetization planes were fit using the methods of *Kirschvink* [1980], and their best intersection was determined using Bingham statistics [*Onstott*, 1980].

The magnetic mineralogy in the rocks was investigated by inspection of 31 polished grain mounts from magnetic separates and of 29 polished thin sections (at least one representative sample from each site) under reflected light at magnifications up to 1000×. Rock chips were also inspected with a scanning electron microscope (SEM) to characterize the sizes, shapes, compositions and paragenetic relations of the magnetic oxide minerals. Semiquantitative chemical analyses of the Fe-Ti oxide minerals were made with an energy dispersive X ray analyzer. Curie temperatures of magnetic separates were measured using an electrobalance in an induction of 180–200 mT at temperatures as high as 625°C in air. An electromagnet was used for acquisition of isothermal remanent magnetization (IRM) in samples in inductions up to 930 mT and for backfield demagnetization to determine coercivity of remanence (Bcr). Measurements of bulk magnetic susceptibility were made on an instrument similar to that described by *Christie and Symons* [1969], with a sensitivity of about 5×10^{-7} SI volume units. Anisotropy of magnetic susceptibility was measured using a 160 cm^3 inducing coil using a 6 or 12 orientation measurement procedure. Samples were given an anhysteretic remanent magnetization (ARM) in a 100-mT peak alternating induction, with the Earth's field as the biasing induction.

Paleomagnetic Results

The natural remanent magnetization (NRM) in all specimens had declinations in the northern hemisphere and moderately inclined positive inclinations. NRM moments varied greatly with geographic location, ranging from as low as 10^{-3} A m^{-1} in the easternmost sites to as high as 2×10^{-1} A m^{-1} in the westernmost sites.

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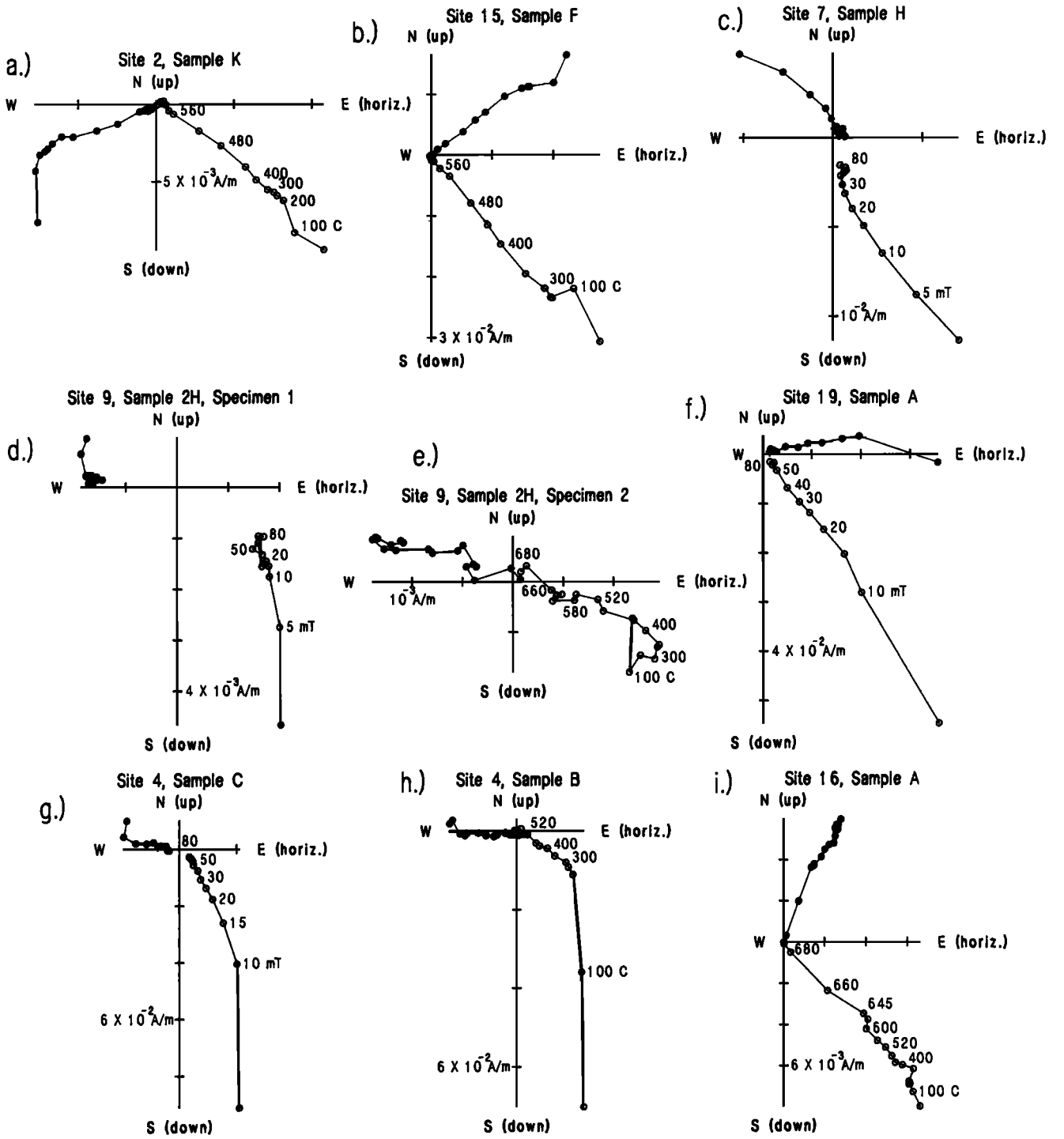


Fig. 2. Orthogonal demagnetization diagrams showing representative behavior of the Preuss samples during thermal and AF demagnetization. Solid and open circles are projections on the horizontal and vertical planes, respectively. Demagnetization levels are given in degrees Celsius (°C) and milliTesla (mT). Descriptions of specific examples are given in the text.

Thermal and alternating field (AF) demagnetization initially removed a component of magnetization at unblocking temperatures to 300°–350°C and at alternating fields to 15–25 mT in most samples. This component was probably a viscous remanent magnetization (VRM).

The characteristic component of magnetization in the Preuss was isolated over unblocking temperatures of 400°C to 560°–600°C. This component was typically represented by a well-defined linear demagnetization path on vector diagrams (Figures 2*a* and 2*b*), although some diagrams show slightly curved demagnetization paths. AF demagnetization was often not as effective as thermal demagnetization in isolating the characteristic magnetization in some eastern sites (e.g., 7, 9, 10). Vector diagrams often showed curved demagnetization paths as the VRM component was being removed (Figure 2*c*). In samples with this behavior, peak

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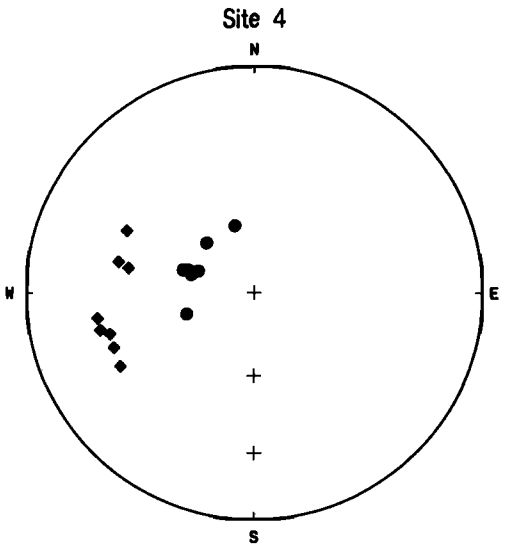


Fig. 3. Equal-area projection illustrating the systematic difference of directions of characteristic magnetizations isolated by AF (circles) and thermal (diamonds) demagnetization at site 4. All points are projections from the lower hemisphere.

inductions of 70–100 mT were not sufficient to remove a component (Figure 2*d*) that could be removed by thermal demagnetization at unblocking temperatures as high as 680° in a different specimen from the same sample (Figure 2*e*). In more western sites (e.g., 4, 12, 13, 15, 18, 19, 20, 25) an apparently well-defined characteristic component was removed at peak inductions between 20 and 70 mT (Figure 2*f*); however, at some sites the direction of this component is systematically different from that defined by thermal demagnetization (Figures 2*g*, 2*h*, and 3).

Directions of characteristic components within most sites are reasonably well grouped. If characteristic components from both thermal and AF demagnetization are combined, values of the precision parameter k range from 2 to 109 for the site means, with most ranging from 15 to 50 (Table 1). If only the components isolated by thermal demagnetization are included, the site means typically have slightly higher values of k. Only the means of components that were defined by thermal demagnetization are used in later analyses because thermal demagnetization was most effective in isolating the characteristic components from components removed at low demagnetization levels. Within a site, directions and demagnetization behavior of the characteristic components did not change systematically with changing stratigraphic position. Three sites (6, 8, 24), each exhibiting atypical features related to deformation or alteration, had highly dispersed characteristic components. Sandstone layers at site 6, which is located in the overturned limb of a fold near the trace of the Absaroka thrust, are offset by small thrust faults, and unlike other sites, adjacent mudstones contain a spaced cleavage. Thus it is possible that the directions of magnetization at this site were, in some manner, dispersed by an atypically high level of strain in the Preuss. Site 8 was confined to a single sandstone layer that had undergone extensive alteration, resulting in bleaching. Samples from this site had unusually low magnetization moments. Sandstone layers at site 24 were highly fractured and cut by numerous small faults that may be related to salt diapirism. The results from these sites are excluded from the following discussion of paleomagnetic results. All the remaining site means have positive inclinations both before and after tilt correction.

Although directions of characteristic components of magnetization are well grouped within a site, the in situ directions of site means are not well grouped (k = 4, Figure 4*a*). Full structural correction for the bedding tilt at each site improves the grouping of site mean directions (k = 12, Figure 4*b*), however, if all the site means are untilted in a series of partial steps, a minimum dispersion (k = 27) comes at 66% unfolding (Figure 4*c*). As noted earlier, minimum dispersion at partial untilting is a geometric trait expected for magnetization components acquired during folding. Data from sites on opposing limbs of the individual folds best illustrate this phenomenon (Figure 5). Directions of components from opposing limbs are most closely grouped at an intermediate stage of the unfolding (ranging from 52 to 74% among the different folds). Complete unfolding causes the site mean directions to cross past each other. For all sites from this study, tectonic correction moves the site directions along small circle paths that pass through or near Late Cretaceous or early Tertiary directions (Figure 6), the expected directions for the area at the time of folding. However, we find that the mean direction from a group of magnetization components at the stage of minimum dispersion is not always coincident with the expected direction at the time of folding (Figure 5*c*). Using the mean direction at minimum dispersion as the orientation of the geomagnetic field during folding, as has been done in other studies of synfolding magnetization [e.g., *McCabe et al.*, 1983; *Kent and Opdyke*, 1985; *Granier et al.*, 1986], could be incorrect.

In the easternmost sites (7, 9, 10, 16, 17, 23), substantial magnetization (>25% of NRM) remained at unblocking temperatures above 600°C in the Preuss. This reflected contributions from hematite. Typically, the component of magnetization carried by hematite was not well defined within a specimen because of erratic demagnetization behavior. Hematite components that were well defined within individual specimens (Figure 2*i*) have directions that are highly dispersed both before and after tilt correction (Figure 7). These components have both positive and negative inclinations. After tilt correction there is a slight decrease in dispersion (k increases from 2 to 3), which suggests that the magnetization carried by hematite may predate at least some stages of folding. Nevertheless, the dispersed directions of components carried by hematite suggest that hematite did not efficiently record a geomagnetic field position.

MAGNETIC MINERALOGY

Identification of the magnetic minerals carrying the characteristic component in the Preuss is key to understanding the origin of the characteristic magnetization. Several rock magnetic properties were investigated and integrated with field and petrographic observations in an attempt to learn the composition, abundance, grain size, and origin of the magnetic minerals. Evidence for both primary and secondary magnetic minerals was sought. Most of the following discussion focuses on properties of magnetite, because the unblocking temperatures (about 400°–600°C) indicate that magnetite is the carrier of the characteristic magnetization in the Preuss.

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TABLE 1. Preuss Sandstone Paleomagnetic Data

	Site		Strike and Dip	Treat	N/N_0	D_0/I_0	a_{95}	k	D_1/I_1	Virtual Geomagnetic Poles	
	Latitude	Longitude								Latitude	Longitude
3P1	43.370	−111.497	125/114	AF+T	18/19	7/18	6.3	31	250/40	2	188
				AF	10/10	7/20	9.8	25	249/38	0	188
				T	8/9	7/16	8.9	40	252/42	4	188
3P2	43.243	−111.237	315/40	AF+T	15/15	248/41	3.7	109	295/72	48	201
				AF	7/7	245/42	7.1	74	292/74	47	206
				T	8/8	250/41	4.1	183	297/70	48	197
3P3	43.386	−111.468	308/63	AF+T	16/17	257/60	8.3	21	10/48	73	37
				AF	7/8	254/64	14.9	17	16/47	70	24
				T	9/9	259/57	11.1	23	5/48	75	50
3P4	43.308	−111.883	335/65	AF+T	15/15	274/52	11.0	13	34/55	63	342
				AF	7/7	295/68	8.7	50	43/38	48	356
				T	8/8	265/37	11.1	26	15/70	76	287
3P6*	43.207	−110.846	340/97	AF+T	14/15	278/38	24.7	4	42/38	49	358
				AF	8/9	300/57	24.6	6	45/14	37	7
				T	6/6	262/6	33.1	5	29/72	67	291
3P7	43.233	−110.785	170/35	AF+T	13/13	337/69	14.6	9	290/46	32	178
				AF	6/6	317/67	20.3	12	285/39	26	176
				T	7/7	356/69	23.7	7	296/52	39	180
3P8*	43.288	−110.803	175/21	AF+T	11/15	28/59	50.6	2	350/64	82	182
				AF	5/6	341/43	31.8	7	325/35	52	131
				T	6/9	104/33	81.1	2	112/53	9	301
3P9	43.147	−110.810	333/58	AF+T	14/15	306/51	17.3	6	15/41	67	32
				AF	6/8	312/55	20.5	12	21/38	62	24
				T	6/7	293/40	14.4	23	359/49	77	73
3P10	43.144	−110.818	148/40	AF+T	14/14	31/46	11.6	13	330/72	67	204
				AF	7/7	33/45	19.0	11	335/73	69	211
				T	7/7	29/47	17.5	13	325/71	65	199
3P12	42.403	−111.005	340/65	AF+T	17/17	319/67	5.2	47	45/31	44	359
				AF	10/10	329/67	7.3	44	45/27	43	2
				T	7/7	305/65	7.0	76	45/37	47	355
3P13	42.392	−110.973	182/51	AF+T	14/14	52/54	6.1	43	320/59	60	168
				AF	7/7	41/59	5.8	111	313/52	52	161
				T	7/7	60/47	8.2	55	332/66	69	183
3P15	42.259	−110.843	175/25	AF+T	13/13	65/45	8.0	28	47/67	57	310
				AF	6/6	74/35	12.9	28	67/59	41	318
				T	7/7	55/52	3.8	250	21/70	72	292
3P16	42.332	−110.508	175/31	AF+T	14/15	36/54	9.2	20	344/63	78	175
				AF	6/7	24/66	11.4	35	315/62	58	260
				T	8/8	42/45	10.8	27	4/60	87	356
3P17	42.655	−110.470	180/40	AF+T	12/13	32/59	10.7	16	323/56	61	161
				AF	7/7	12/64	7.3	69	311/48	49	157
				T	5/6	55/55	16.8	22	333/68	70	195
5P18	42.177	−110.940	338/67	AF+T	19/21	307/53	10.5	11	28/37	58	13
				AF	9/10	316/62	15.9	12	38/30	48	6
				T	10/11	296/44	10.0	24	18/46	69	18
5P19	42.165	−110.920	182/93	AF+T	17/20	43/53	15.7	6	301/21	30	146
				AF	10/10	9/66	15.6	11	296/0	19	141
				T	7/10	65/28	12.2	25	311/49	49	159
5P20	42.349	−110.973	250/17	AF+T	19/20	11/73	4.0	70	356/57	84	102
				AF	10/10	10/70	6.5	57	357/54	82	86
				T	9/10	12/76	4.8	114	355/60	86	143
5P23	42.066	−110.475	177/18	AF+T	17/19	28/50	15.5	6	5/56	83	34
				AF	9/9	39/34	25.0	5	27/44	63	7
				T	8/10	8/63	11.7	23	333/61	70	168
5P24*	40.777	−111.708	210/97	AF+T	20/20	142/39	56.3	1	278/37	20	169
				AF	10/10	174/59	17.0	9	275/9	7	158
				T	10/10	61/−60	53.4	2	92/24	6	327
5P25	40.770	−111.761	57/65	AF+T	18/18	339/38	11.6	10	110/74	26	280
				AF	9/9	356/50	10.1	27	113/56	9	297
				T	9/9	328/23	14.8	13	338/88	44	246
Mean of 17 site means after thermal demagnetization											
In situ						351/64	19.3	4			
Fully untilted						344/66	10.8	12			
66% untilted						347/67	6.9	27			

Paleomagnetic data from Preuss Sandstone, Wyoming-Idaho-Utah thrust belt. Positive latitudes and longitudes are north and east for locations of sites and virtual geomagnetic poles. Strike and dip of bedding is at site. Dip direction is 90° clockwise from strike. Dips > 90° represent overturned beds. Treat, alternating field (AF) or thermal (T) demagnetization treatment; N/N_0, number of samples accepted/number of samples demagnetized; D_0/I_0 and D_1/I_1, in situ and tilt-corrected site mean declination/inclination; a_{95}, 95% confidence cone about mean direction; k, precision parameter [*Fisher*, 1953].

*Denotes sites not used for calculation of means of site level directions.

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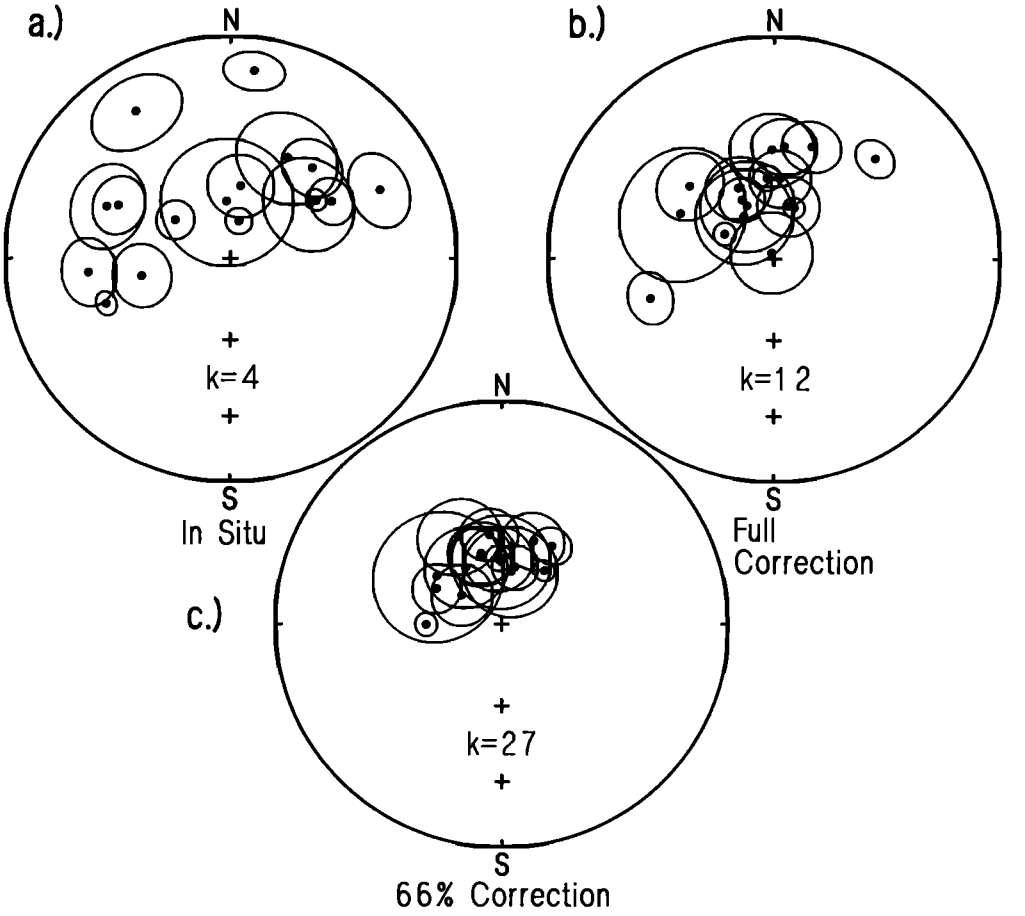


Fig. 4. Equal-area projections showing site mean directions of Preuss magnetizations isolated by thermal demagnetization and their respective cones of 95% confidence. Site mean directions are shown (*a*) in situ, (*b*) after full tilt correction, and (*c*) after partial tilt correction. The precision parameter *k* [*Fisher*, 1953] is greatest at 66% untilting.

Initial evidence that primary detrital minerals would play an important role in the rock magnetic properties of the Preuss came from field observations. Heavy mineral laminations are common in sandstones from the Preuss in the western parts of the thrust belt, and highly magnetic minerals were easily separated from the laminations at the outcrop. Eastward, the occurrence of the heavy mineral laminations in the Preuss becomes less frequent.

Petrography

Petrographic inspection showed that the Preuss contains well-rounded Fe-Ti oxide grains, including magnetite, distributed throughout the rock and concentrated along heavy mineral laminations (Figure 8*a*). These grains have similar shapes and are typically slightly smaller than the silicate framework grains of the sandstones. Thin section examination also revealed evidence for tectonic strain in silicate and magnetite grains.

Magnetite occurs in two predominant grain types: as optically homogeneous grains and as subdivided grains. Well-rounded grains (20–100 μm diameter) of optically homogeneous magnetite are the predominant form (55–90% in individual samples) of magnetite in the Preuss (Figure 8*b*). Many of these grains have been partly oxidized to maghemite and hematite adjacent to rims and internal cracks. No petrographic textures were found that would suggest that homogeneous magnetite replaced a preexisting mineral (e.g., magnetite rims on sulfides or on hematite). Subdivided grains comprise from 10 to 45% of the total population of magnetite grains in individual samples. In the subdivided grains (Figure 8*c*), magnetite occurs as numerous small (<20 μm) blocks between trellis-patterned laths of ilmenite, spinel, and hematite. The textures and compositions of the Fe-Ti oxides in the subdivided grains are characteristic of those produced by high-temperature oxidation-exsolution of ulvospinel-magnetite solid solutions in igneous rocks at temperatures above 400°C [*Ramdohr*, 1980]. These textures and mineralogies must have formed before the grains were deposited in the Preuss. Modal analyses [*Fishman et al.*, 1989] indicate that both the total magnetite content and the proportion of magnetite in the heavy mineral population is greater in the western sites than in the eastern sites. No secondary magnetite occurring as or within interstitial cement of the framework grains was observed petrographically.

In addition to the magnetite grains, many other forms of Fe-Ti oxide minerals are common in the Preuss. Detrital grains of the ilmenite-hematite series (monocrystalline and polycrystalline specular hematite, martite, ilmenite, and intergrowths of ilmenite and hematite) are dominant in the heavy mineral population (Figures 8*d* and 8*e*). High-temperature exsolution in magmatic source rocks produced

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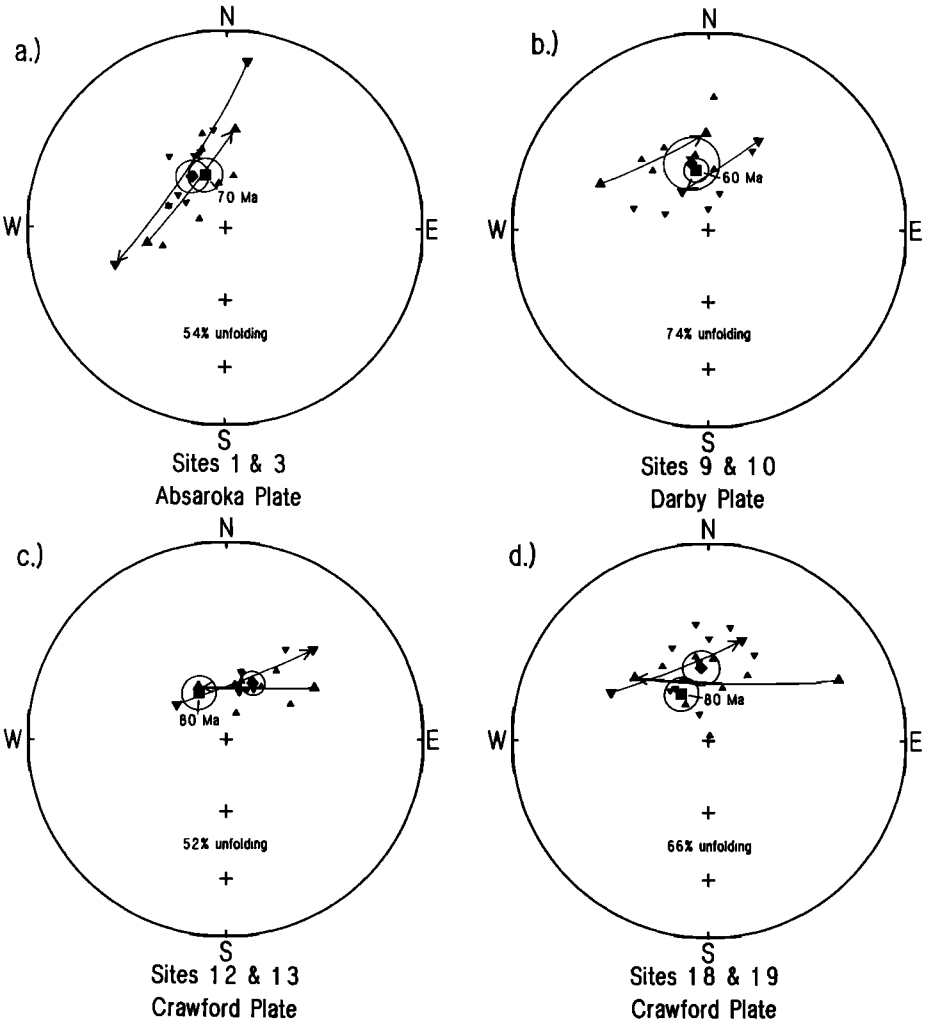


Fig. 5. Equal-area projections of directions of characteristic magnetization isolated by thermal demagnetization from each of four individual folds after partial unfolding. Sample directions from opposing fold limbs are shown by small triangles with points in opposing directions. Small circle paths connect in situ and fully untilted site means (large triangles) for the fold limbs. Large diamonds show mean directions and their cones of 95% confidence of the sample magnetization components from the individual folds at the stage of minimum dispersion during untilting. Large squares show the expected direction at the approximate time of movement of the thrust plates on which the folds occur (calculated from *Irving and Irving* [1982]).

the ilmenite-hematite intergrowths [*Ramdohr*, 1980]. Pseudobrookite and ilmenorutile, products of high-temperature oxidation in source rocks, are also common in the Preuss.

The diagenetic history of the Preuss has been investigated and summarized by *Fishman et al.* [1989]. The only authigenic Fe-oxide observed in the Preuss is a thin ferric oxide coating on most of the framework grains. The ferric oxide precipitated during earliest phases of the diagenesis of the Preuss and survived later diagenetic alterations. The presence of diverse Fe-Ti oxides of varying oxidation states in the heavy mineral laminations and the preservation of the ferric oxide coating on the framework grains suggest that pore waters in the Preuss never promoted a pervasive dissolution, reduction, or oxidation of the Fe-Ti oxides. This and the absence of sulfide minerals indicate that the Preuss has not experienced sulfidic-reducing conditions common in areas of hydrocarbon seepage [*Fishman et al.*, 1989]. Abundant calcite cement, which sealed most of the pore spaces before the final stages of deformation [*Fishman et al.*, 1989], promoted preservation of the magnetite throughout the postdeformation history of the Preuss. Marine carbon isotopic compositions for the calcite cement [*Fishman et al.*, 1989] preclude a possible hydrocarbon carbon source for the carbonate diagenesis. On the basis of the shape, size, and distribution of the magnetite grains and on the chemical conditions in which diagenetic alterations occurred, we conclude that both the optically homogeneous and the subdivided magnetite grains in the Preuss were deposited as detrital grains and that they underwent little or no postdepositional alteration.

Framework grains in the Preuss were strained during tectonic deformation [*Fishman et al.*, 1989]. Many lithic and TiO_2 (anatase or rutile) grains have been indented by more competent quartz or feldspar grains. Moreover, quartz grains commonly display undulatory extinction, and calcite, as cement and as a fracture filling that crosscuts framework grains, displays twinning. The undulatory extinction in quartz and the twinning in calcite probably resulted from intracrystalline tectonic strain. In samples from several sites

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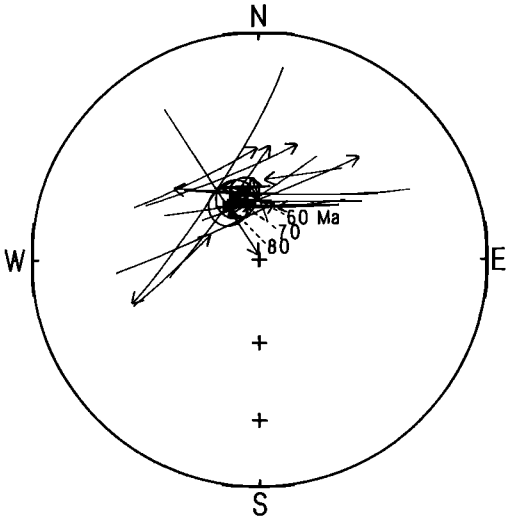


Fig. 6. Equal-area projection showing small-circle paths that the site mean directions follow upon full correction for bedding tilt. The 60, 70, and 80 Ma expected directions for the thrust belt [from *Irving and Irving*, 1982] correspond to approximate ages of major movement of the Darby-Hogsback, Absaroka, and Crawford-Meade thrust plates, respectively [*Wiltschko and Dorr*, 1983].

some homogeneous magnetite grains display brittle fractures. Locally, these grains appear to have been expelled by cataclastic flow from between other, apparently unaffected, grains (Figure 8*f*). Typically, the proportion of homogeneous magnetite grains in an individual sample that are cataclastically deformed is low (<5% of the magnetite population). Cataclasis appears to have been preferentially developed at discrete points of contact between adjacent grains which focused concentrated stress levels. Cataclastic textures were not well developed in the subdivided magnetite grains or other Fe-Ti oxides grains, suggesting that the rheological properties of the homogeneous magnetites must differ from those of the other oxide minerals.

Rock chips of the Preuss were also inspected using a SEM at magnifications as high as 30,000×. A search was undertaken for small magnetite grains within the interstitial calcite cement and for magnetite coats or rims on the framework grains that might be diagenetic; none were observed. Using energy dispersive analysis (EDA), semiquantitative chemical compositions were determined for the two populations of detrital magnetite grains. Both grain types contain titanium in addition to iron. Weight percent of TiO_2, estimated from magnetite blocks in subdivided grains, ranged from 2.5 to 7.6, whereas that estimated from the homogeneous grains ranged from 2.3 to 6.1. These Ti contents correspond to about 7–22 and about 7–18 mol % ulvospinel in the subdivided and homogeneous grains, respectively.

Rock Magnetism

Rock magnetic experiments provided diverse information bearing on the origin of the magnetite and on the origin of remanence in the Preuss. Constraints on the composition of magnetite and on the relative distributions of magnetite and hematite were derived from thermomagnetic analysis and from acquisition and removal of isothermal remanent magnetization (IRM). Magnetite content was inferred from measurements of magnetic susceptibility and moments of IRM. Information on the relative size of magnetite particles was interpreted from comparison of values of magnetic susceptibility and moments of anhysteritic remanent magnetization. Anisotropy of magnetic susceptibility was used mainly to test for the influences of tectonic strains and sedimentation on the magnetic fabric.

Thermomagnetic analysis. To compare with the EDA method, magnetite compositions were also estimated by thermomagnetic analysis from which Curie temperatures were measured. Magnetic separations removed black, highly magnetic grains from the Preuss that were of similar size and shape as the detrital magnetite grains observed petrographically. Thermomagnetic curves from these grains (Figure 9) show Curie temperatures ranging from 531° to 562°C, with a mean of 553°C (standard deviation = 7° from 17 samples from 17 different sites). The Curie temperatures correspond to about 4–9 mol % ulvospinel in the titanomagnetite series. These compositions are similar to, although slightly lower than, compositions determined for the detrital magnetites by

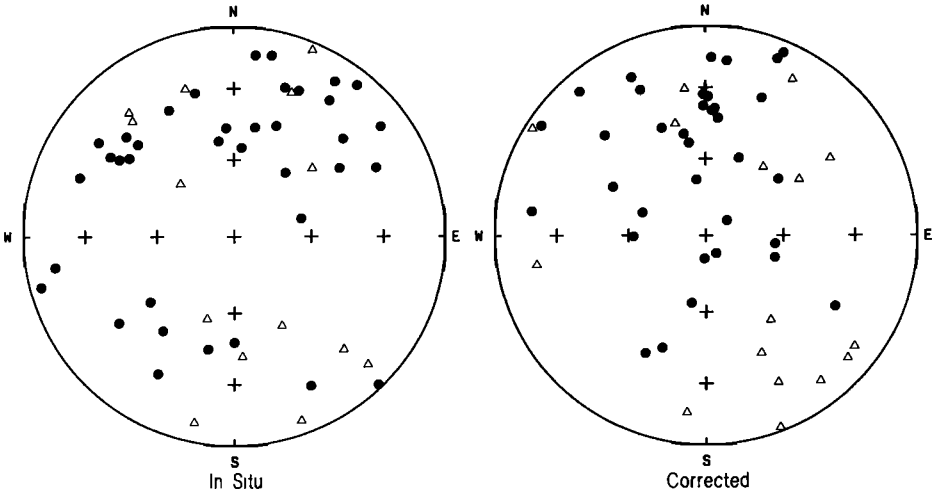


Fig. 7. Equal-area projection showing the in situ and tilt-corrected distribution of all hematite components isolated at demagnetization temperatures >600°C in the Preuss. Closed and open circles are projections from the lower and upper hemispheres, respectively.

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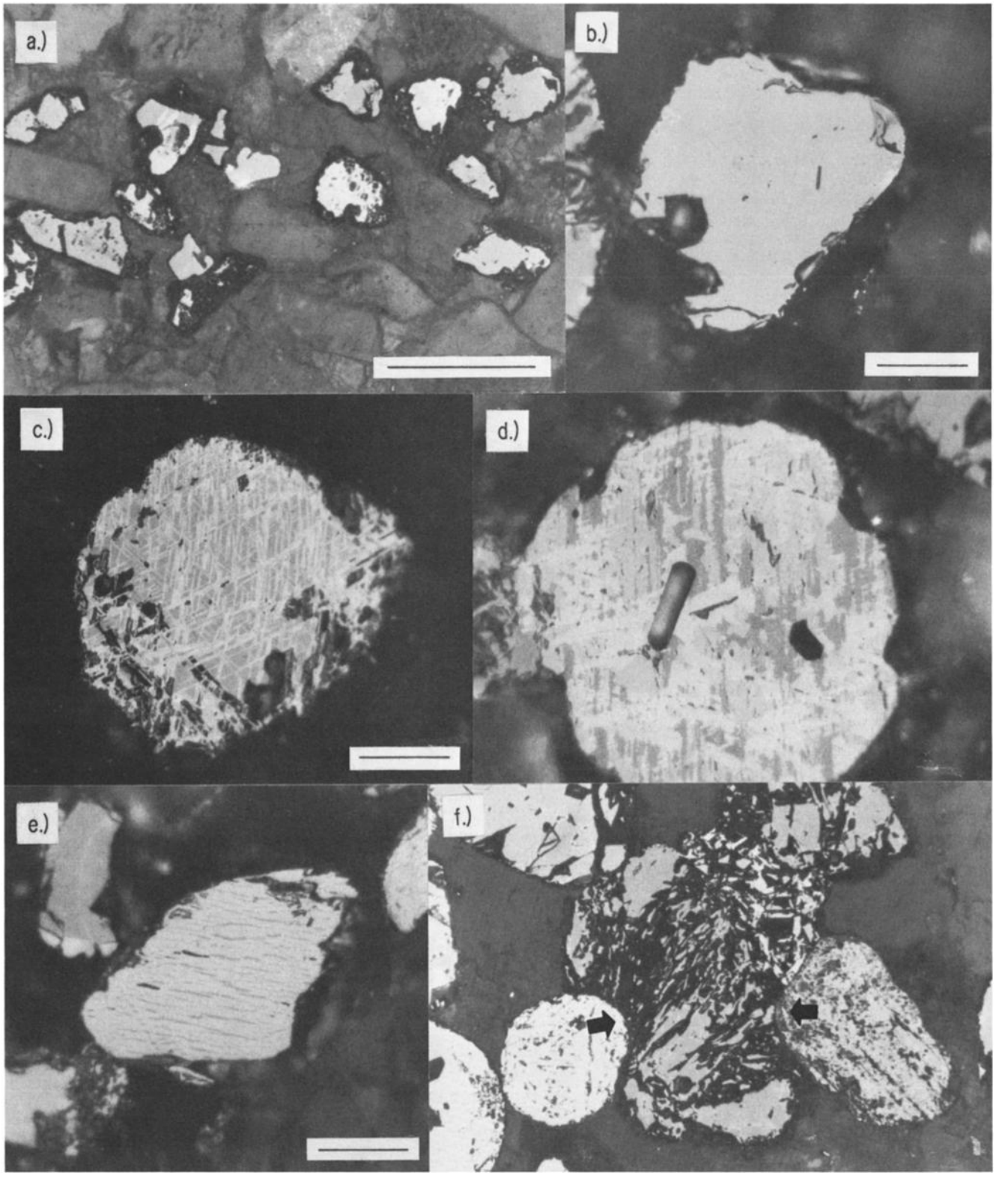


Fig. 8. Photomicrographs of detrital Fe-Ti oxide minerals from the Preuss. (*a*) Concentration of Fe-Ti oxide minerals (dark grains) in a heavy mineral lamination. Bar is 200 μm. (*b*) Homogeneous magnetite grain with rims and internal cracks replaced by maghemite and/or hematite (lighter areas). Bar is 100 μm. (*c*) Subdivided grain with magnetite (dark) divided by laths of hematite (light). Bar is 100 μm. (*d*) Polycrystalline hematite grain viewed through crossed nichols. Bar is 100 μm. (*e*) Grain composed of ilmenite (dark) and hematite (light) intergrowths. Bar is 50 μm. (*f*) Homogeneous magnetite grains that have been broken by brittle fracture (see arrows) from a heavy mineral lamination. Bar is 200 μm.

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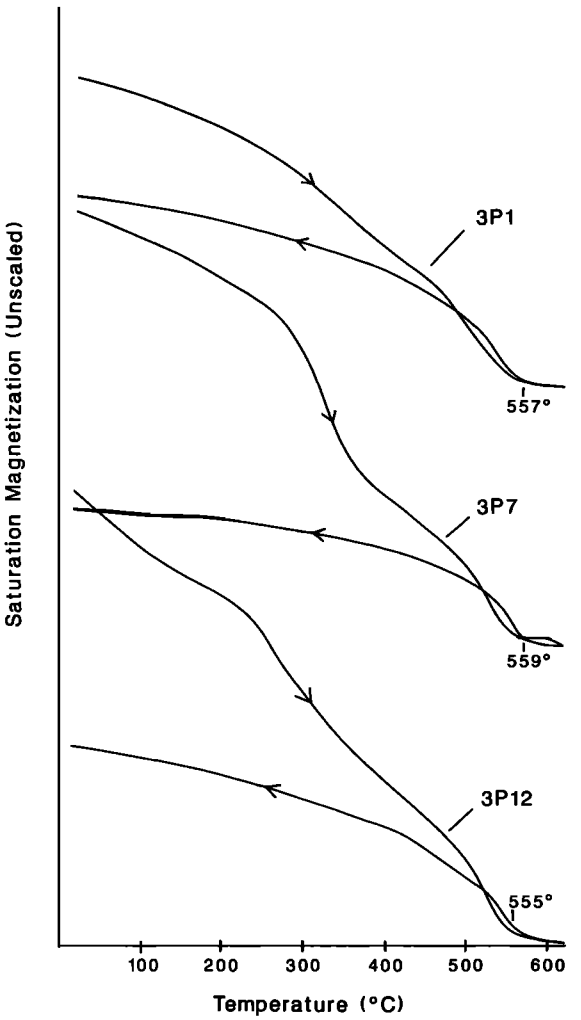


Fig. 9. Thermomagnetic curves of magnetic separates from representative Preuss samples. All analyses were conducted in air. Curie temperatures indicate that the magnetic carrier is low-Ti titanomagnetite. Step-like magnetization loss at 250°–400°C in some heating curves reflects the presence of maghemite.

EDA methods. Also common in the thermomagnetic curves was a loss of magnetization during heating between 300° and 400°C that was not reversible during cooling. This behavior is characteristic of maghemite and is consistent with our petrographic observations of maghemite that commonly rims the homogeneous magnetite grains. Because Curie temperatures were determined for magnetic separates, we were aware that any fine-grained diagenetic magnetite embedded in the matrix of the Preuss might not be sampled. We performed thermomagnetic measurements of rock chips to compare any differences between the thermomagnetic characteristics of the bulk rock and magnetic separates, but unfortunately a high-temperature (500°–620°) weight loss in the rock chips masked the loss of bulk magnetization, precluding the comparison.

Isothermal remanent magnetization acquisition. Acquisition of IRM for 34 representative samples (at least one per site) of the Preuss indicated that magnetite is the dominant carrier of the magnetization, but the measurements also highlighted a geographic variation of magnetic hysteresis properties. In samples from all sites the highest rate of IRM acquisition occurred at inductions from 0 to 200–300 mT (Figure 10*a*), as is characteristic of an inverse spinel phase such as magnetite. The amount of remanence acquired at inductions >300 mT varied greatly and systematically among samples from western and eastern sites across the thrust belt. Samples from eastern sites acquired as much as 40% additional remanence at inductions from 300 to 930 mT, reflecting large contributions from hematite. Samples from western sites typically acquired only 2–4% additional remanence at the same inductions. This, along with an order of magnitude increase in the moment of IRM at 930 mT, reflects both a higher absolute amount of magnetite grains and a higher proportion of magnetite grains to hematite grains in western sites as compared to the eastern sites. These conclusions are consistent with modal analyses from petrographic studies of the Preuss [*Fishman et al.*, 1989]. The geographic variation in relative abundances of magnetite and hematite is also apparent in the range of coercivities of remanence determined by backfield demagnetization (Figure 10*b*). In westermost sites (Meade and Crawford thrust plates), Bcr values range from 21 to 33 mT, whereas in easternmost sites (Darby-Hogsback thrust plate) minimum Bcr values are 58 to 242 mT. Bcr values from sites on the Absaroka plate are intermediate, 40–74 mT, with the exception of lower values (25 and 28 mT) from the southernmost sites, 24 and 25. These later sites are west and distant from the remainder of sites on the Absaroka plate.

Magnetic susceptibility. Magnetic susceptibility of the Preuss varies greatly (2.7×10^{-5} to 8.7×10^{-2} SI volume units) and is a measure of magnetite abundance. Regionally, the average site values of magnetic susceptibility change systematically across the thrust belt, from highest values in the west to lowest values in the east (Figure 11). This change corresponds with both a decrease in the modal abundance of detrital magnetite grains [*Fishman et al.*, 1989] and the decrease in the abundance of heavy mineral laminations in the Preuss. Specimens containing heavy mineral laminations have susceptibilities that are higher by an order of magnitude than susceptibilities from the remaining samples at an individual site. These observations indicate that the abundance of detrital magnetite grains controls the variation in magnetic susceptibility in the Preuss. We interpret the decrease of magnetic susceptibility across the thrust belt to reflect primarily an original and gradual decrease in the amount of detrital magnetite that reached depositional sites toward the east. Both *Hileman* [1973] and *Jordan* [1985] indicate that the major source area for the Preuss lay to the west. This gradient of susceptibility was accentuated by later thrusting and folding, which shortened the area by about 50% [*Dixon*, 1982].

That the magnetic susceptibility of the Preuss appears to be controlled by the abundance of detrital magnetite does not alone preclude the presence of fine-grained diagenetic magnetite. Experimental results indicate that large multidomain grains (>15–20 μm) have greater initial susceptibilities than pseudosingle domain (about 0.1–15 μm) or single domain (about 0.03–0.1 μm) grains of magnetite by as much as a factor of 2, if the grains have had a similar preparation history [*Dunlop*, 1986; *Heider et al.*, 1987]. Thus large detrital grains might still control the magnetic susceptibility even if an undetected population of fine-grained secondary magnetite is present in the Preuss. Experimental data, however, have also highlighted the dependence of initial susceptibility on preparation history [*Dunlop*, 1986]. If secondary

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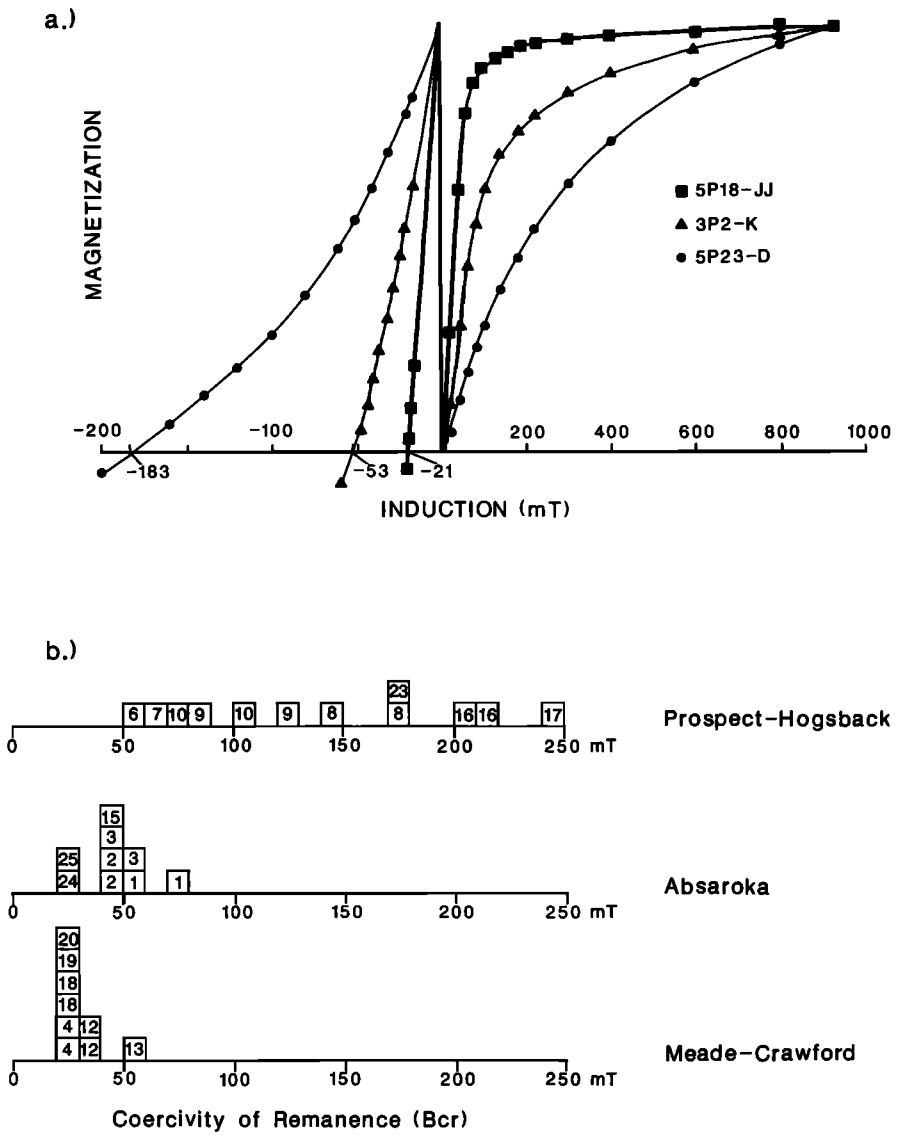


Fig. 10. (*a*) Acquisition of IRM and backfield demagnetization for Preuss samples. Variation of behavior reflects increase in hematite content in samples from western (18) to intermediate (2) to eastern (23) sites. Magnetization moments at 930 mT induction are 2.41 A m^{-1} (5P18-JJ), 1.17 A m^{-1} (3P2-K), and 0.29 A m^{-1} (5P23-D). Coercivity of remanence (Bcr) values for the samples are given at the intersection of the backfield demagnetization curves with the zero magnetization axis. (*b*) Histograms of Bcr values from samples from the eastern Darby and Hogsback, intermediate Absaroka, and western Meade and Crawford thrust plates. The sites from which individual samples were taken are given in individual blocks. Higher Bcr values reflect increasing content of hematite.

single-domain magnetite experienced a different history, it could have greater initial susceptibility than multidomain detrital magnetite.

Anhysteretic remanent magnetization. Plots of anhysteretic remanent magnetization (ARM) (versus magnetic susceptibility (MS) were made to search for systematic variations of detrital magnetite grain size and to test for the possible presence of fine-grained secondary magnetite. Following the work of *Banerjee et al.* [1981], the ratio of ARM to MS will be lowest in coarse-grained magnetite and will increase gradually for finer grain sizes. Direct comparison of ARM/MS ratios of the Preuss samples with ratios for sized fractions of magnetite of *Dankers* [1978] (shown by *Banerjee et al.* [1981]) is not possible because of differences in the manner in which the ARM was imparted. Nevertheless, the relative changes of ARM/MS among the Preuss samples and sites are useful. Analysis of specimens within a site typically produced data that fell on a common line (Figure 12), but the slope of these lines varies among the sites. This suggests that there are changes of the size of magnetite grains among the sites. Data from a few of the most weakly magnetic eastern sites (8, 17, 23) do not lie in a well-developed line. The nonlinearity probably reflects poor instrumental resolution of the ARM and MS values rather than a true variation of the magnetite size within the site. Results from these sites are excluded in the following discussion. ARM/MS slopes for individual sites do not vary consistently with geographic position (Figure 13). If the ARM was controlled by detrital magnetite, the variation of ARM/MS slopes may reflect changes of grain size related to changes in depositional environment of the Preuss. A general association of lower ARM/MS slopes (i.e., coarser magnetic grain size) with

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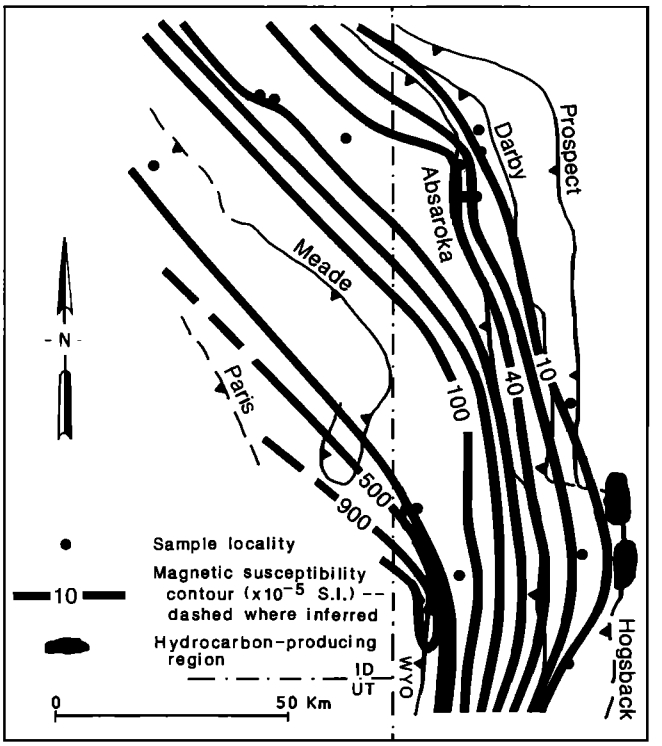


Fig. 11. Contours of site mean magnetic susceptibility (SI volume units) for the Preuss in the northern part of the Wyoming-Idaho-Utah thrust belt. The systematic eastward decrease of magnetic susceptibility is attributed to a decrease in the amount of detrital magnetite shed from an original western source area, although this gradient has been steepened by later structural shortening.

coarser framework grain size of the samples (Figure 14) is consistent with this interpretation. Of course, the effective grain size of individual magnetite blocks within subdivided and fractured grains will be less than the overall grain size.

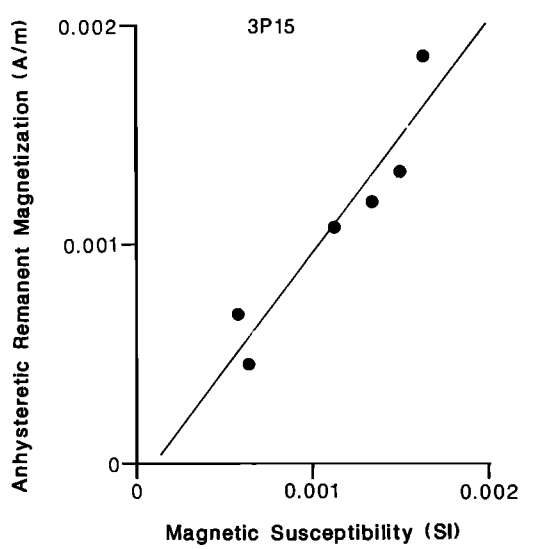


Fig. 12. Plots of anhysteretic remanent magnetization against magnetic susceptibility for samples from a representative site (15) of the Preuss Sandstone. The high correlation (correlation coefficient $R = 0.94$) between the values suggest that the grain size of magnetic minerals in the samples is relatively constant within the site. Best fit line through the data was obtained by linear regression.

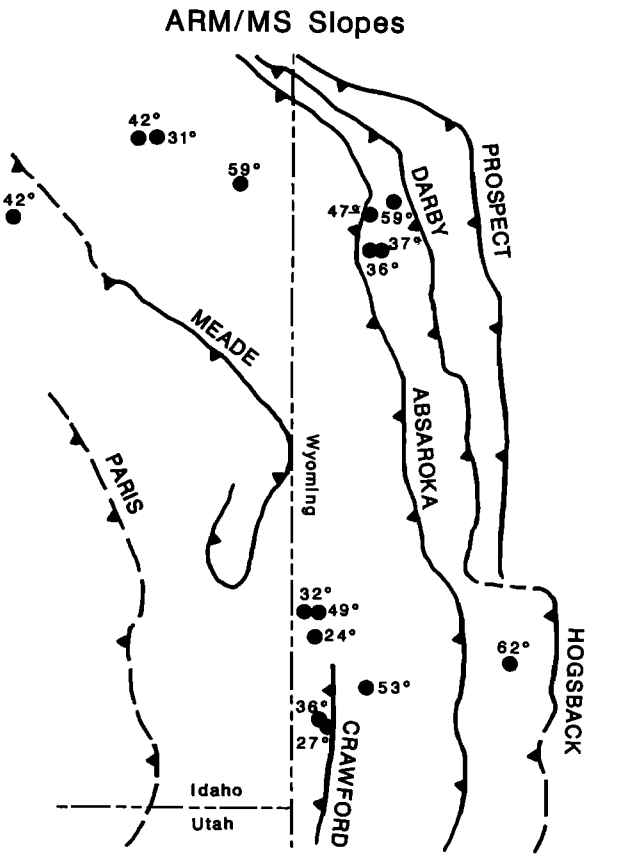


Fig. 13. Geographic distribution of the slopes defined by best fit lines from plots of anhysteretic remanent magnetization (ARM) against magnetic susceptibility (MS), as illustrated in Figure 12, for sites in the northern portions of the thrust belt. The variation of slopes reflects variations of magnetic grain size among the sites.

This may explain some of the scatter of the ARM/MS slopes on the histograms of Figure 14.

Anisotropy of magnetic susceptibility. The anisotropy of magnetic susceptibility (AMS) was measured for samples at all sites except those easternmost sites with lowest bulk susceptibilities (7, 8, 16, 23) to assess the effects of tectonic strain on the magnetite population [*Graham*, 1954]. The details of these data and their structural interpretations will be presented elsewhere, but several general conclusions that bear on the interpretation of the remanence are discussed below. The data also reveal individual specimens in which magnetic anisotropy has been sufficient to affect the remanent directions.

The variation of the shapes of AMS ellipsoids and the directions of the principal susceptibility axes suggest that magnetic fabrics in the Preuss are a composite of an original sedimentary fabric that has been overprinted by a later structural fabric [*Hudson*, 1986]. In all sites the axes of maximum susceptibility, K1, are well grouped, are shallowly inclined, and have declinations that are parallel to the local strike of bedding. At sites retaining a predominantly sedimentary fabric the magnetic foliation (K1–K2 plane) is parallel to the bedding, and the AMS ellipsoids have predominantly oblate shapes (Figure 15*a*). At other sites the magnetic foliations are nearly perpendicular to the bedding, and the susceptibility ellipsoids are predominantly prolate (Figure 15*b*), suggesting that tectonic strain has strongly influenced the petrofabric at these sites. Most sites show

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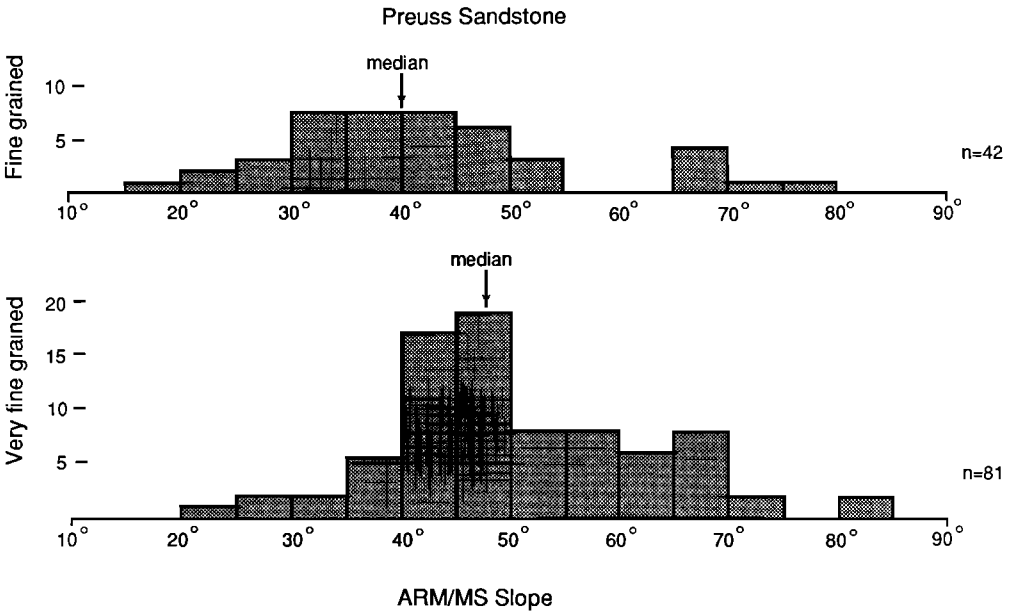


Fig. 14. Histograms of ARM/MS slopes for individual sandstone specimens with fine (125–250 μm) and very fine (62–125 μm) grain size of framework grains. ARM/MS slopes are, on average, lower for fine-grained sandstones than for very fine grained sandstones. This indicates that the magnetic grains carrying the ARM are smaller in the finer-grained sandstones. This relation suggests that depositional processes controlling the grain sizes of the sandstones also influenced the grain sizes of the dominant magnetic minerals in the samples.

combinations of these characteristics (Figure 15*c*). The change from bedding-parallel to bedding-normal foliations concurrent with a change from oblate to prolate ellipsoids and development of a subhorizontal lineation perpendicular to regional compression are all characteristics expected when an original oblate sedimentary fabric has been overprinted by tectonic strain in which shortening is directed parallel to layering [e.g., *Ramsay and Huber*, 1983, pp. 185–188]. Strain analyses of rocks from numerous foreland fold and thrust belts indicate that layer-parallel shortening is a common strain process [e.g., *Nickelsen*, 1979; *Allmendinger*, 1982; *Cooper et al.*, 1983; *Wiltschko et al.*, 1985]. This component of strain probably occurred before the rocks were strongly tilted (i.e., before major folding and thrusting) in response to a subhorizontal regional greatest principal stress field. We interpret the AMS fabrics and their pattern of change to indicate that most of the tectonic strain within sandstones of the Preuss was also produced by horizontal, layer-parallel shortening prior to folding. Similar strain geometries and histories have been documented for other units in the Wyoming-Idaho-Utah thrust belt [e.g., *Allmendinger*, 1982], including the Jurassic Twin Creek Limestone which directly underlies the Preuss [*Mitra et al.*, 1984; *Mitra and Yonkee*, 1985; *Evans and Craddock*, 1985].

The degree of AMS ((K1/K3-1) × 100) in most Preuss samples is low, ranging between 1 and 5%, but samples containing prominent heavy mineral laminations have anisotropies as high as 50% and susceptibility ellipsoids that are strongly oblate. These abnormally high anisotropies almost certainly reflect strong interactions among closely spaced magnetite grains within the heavy mineral laminations. Specimens containing heavy mineral laminations have magnetic foliations that parallel bedding even at those sites where foliations from the remainder of the specimens are perpendicular to bedding. The correlation of AMS fabrics with sedimentary structures further illustrates that magnetic susceptibility of the Preuss is controlled by the distribution of detrital magnetite grains.

The degree of AMS in most specimens of the Preuss is too low to have significantly deflected the remanent directions away from the geomagnetic field position during acquisition of their remanence. However, a deflection of remanence is found in specimens containing prominent heavy mineral laminations. Characteristic components isolated during thermal demagnetization from samples with prominent heavy mineral laminations are typically skewed from the remainder of components from the site toward the site mean axis of maximum susceptibility (Figure 16). This suggests that intergrain interactions among closely spaced detrital grains, which are believed to be responsible for the unusually high anisotropies in these samples, also affected acquisition of the characteristic magnetizations. This indicates that the detrital magnetite grains in the heavy mineral laminations are also the carriers of the characteristic magnetization in these samples.

Detrital Magnetite: Carrier of Characteristic Magnetization

Petrographic and rock magnetic data, constraints on the geochemical nature of fluids during the diagenetic history [*Fishman et al.*, 1989], and the systematic deflection of directions of characteristic magnetization in highly anisotropic specimens which contain heavy mineral laminations are all points that strongly suggest that detrital, Ti-bearing magnetite is the sole carrier of the characteristic magnetization in the Preuss. The possibility that very small particles of secondary magnetite, having single or small pseudosingle domain structure (0.01–0.1 μm diameter), might be present but were undetected by petrographic and SEM examination was considered carefully. This possibility is not supported by any data discussed above.

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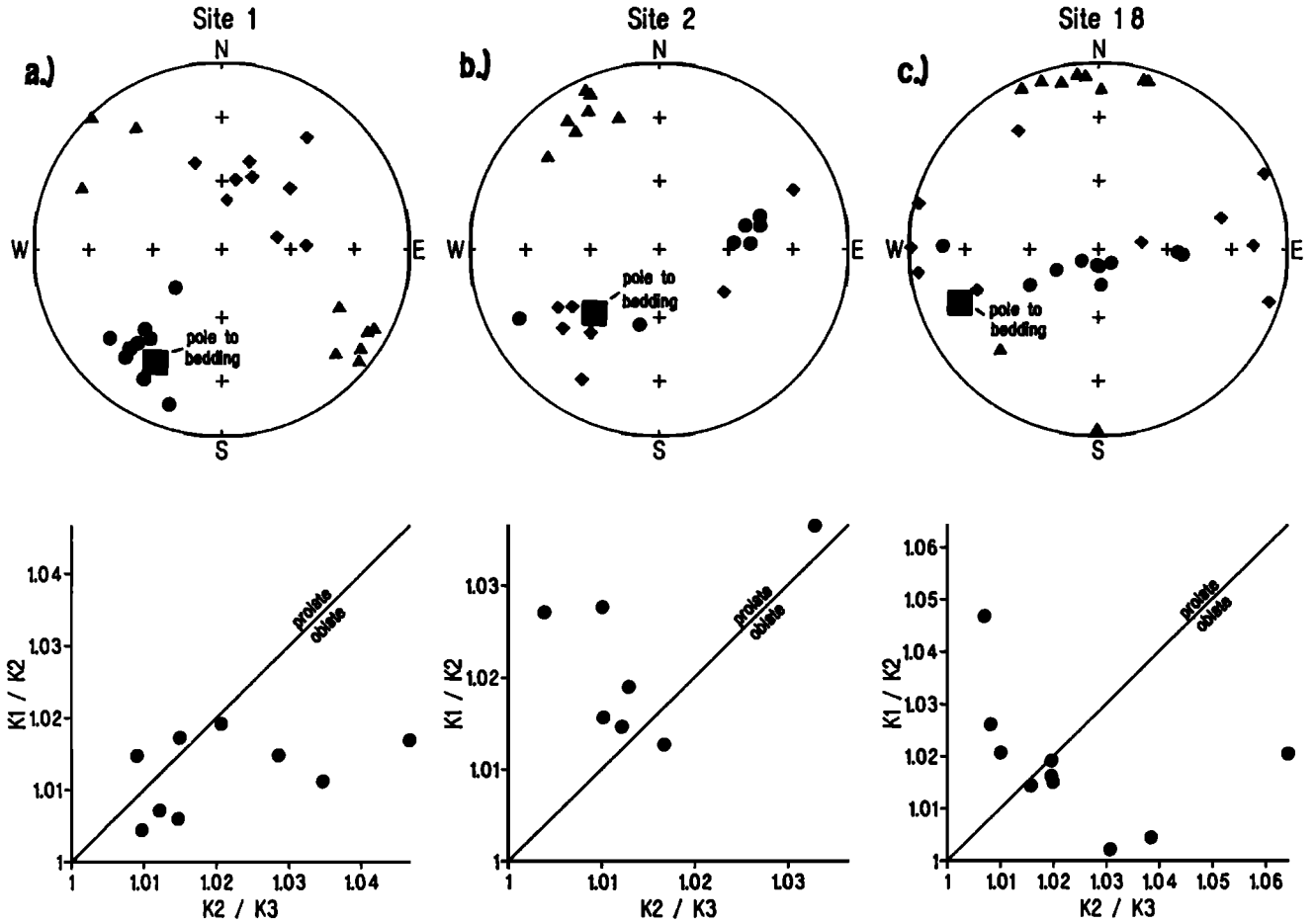


Fig. 15. Plots showing character of anisotropy of magnetic susceptibility of the Preuss samples. Equal-area projections show the axes of maximum (K1), intermediate (K2), and minimum (K3) susceptibility for specimens as triangles, diamonds, and circles, respectively. The pole to bedding for the individual sites are also shown. Plots of K1/K2 versus K2/K3 illustrate the variation of shape of the susceptibility ellipsoid that occurs among the sites. The changes of the shape and orientation, with respect to bedding, of the susceptibility ellipsoids indicate that the magnetic fabric of the Preuss is a composite of an original sedimentary fabric and a tectonic fabric in which most shortening was probably directed parallel to bedding.

Additional evidence that only detrital magnetite grains carry the characteristic magnetization in the Preuss comes from relations among site mean magnetic susceptibility values and site mean moments of NRM and characteristic magnetization. As noted earlier, the correlation of magnetic susceptibility with detrital magnetite abundance and the preservation of sedimentary AMS fabrics indicate that detrital magnetite controls the magnetic susceptibility of the Preuss. However, saturation remanence may be one to two orders of magnitude greater for single or small pseudosingle domain magnetite than for large multidomain grains [*Dunlop*, 1986]. Thus a small amount of secondary, fine-grained magnetite might strongly influence the remanence without noticeably affecting the magnetic susceptibility. It is unlikely that the abundance of such secondary magnetite in the Preuss would vary as the abundance of detrital magnetite. If secondary magnetite strongly influenced remanence therefore, a correlation between moment of remanence and magnetic susceptibility would be most unlikely. A strong correlation is found between the site means of NRM moment and magnetic susceptibility over three orders of magnitude (Figure 17). A strong correlation is also found between the site means of characteristic component moment (as estimated by vector subtraction of the characteristic component) and magnetic susceptibility for the subset of specimens that were subjected to thermal demagnetization. These relations further argue that only detrital grains carry remanence in the Preuss.

Discussion

The term "synfolding magnetization," as commonly used, refers to the observation that directions of magnetization are most closely grouped at an intermediate stage of untilting. However, this term implies that a secondary magnetization was acquired during folding. Acquisition of a secondary magnetization during folding is not the only way to produce this geometry of magnetization direction. To avoid the genetic term "synfolding," we refer to this geometric trait as a "crossover" geometry of magnetization.

Mechanisms That Produce a Crossover Geometry

Mechanisms proposed by other authors and additional mechanisms that could produce a crossover geometry of magnetization are listed in Figure 18. These mechanisms can be subdivided into three major classes: (1) acquisition of

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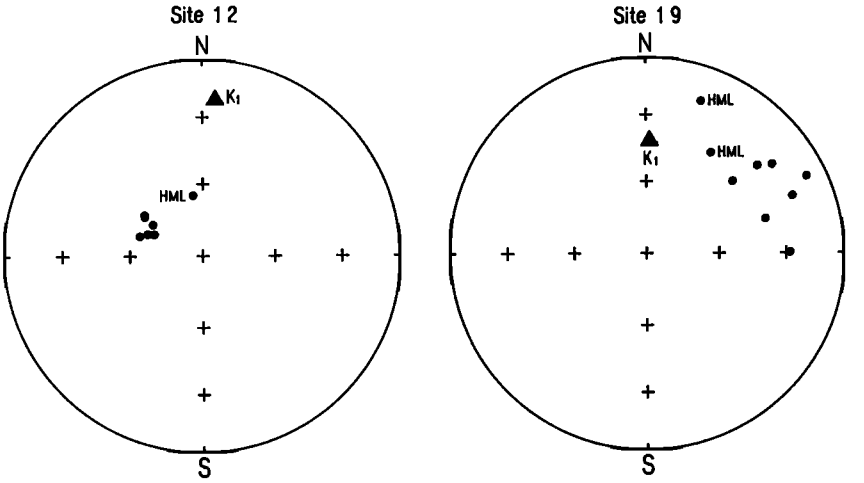


Fig. 16. Equal-area projections illustrating the deflection of characteristic magnetization toward axis of maximum susceptibility in samples containing prominent heavy mineral laminations. Triangles show the orientation of mean K1 axes for the sites.

magnetization during folding (true synfolding magnetization), (2) combinations of prefolding, synfolding, and (or) postfolding components of magnetization, and (3) structural modifications of rocks that contain prefolding magnetization.

The class of "true" synfolding magnetization refers to characteristic magnetization that is entirely secondary and acquired at an intermediate stage of folding by one or more of several possible mechanisms (e.g., CRM, TCRM, VpTRM). For the secondary component to be entirely synfolding any prefolding remanence in the rock must be negligible or have been completely reset within the unblocking temperature or coercivity spectra over which the characteristic magnetization is isolated. The direction of a true synfolding magnetization should be parallel to the orientation of the geomagnetic field at the stage of folding during which remagnetization occurred.

The second class of mechanisms that might cause a crossover geometry of magnetization involves combinations of more than one component of magnetization (Figure 18). For example, a tilted prefolding magnetization combined with a postfolding magnetization can produce a resultant vector with an orientation like that of a true synfolding magnetization. A crossover geometry of a resultant magnetization will be most effectively produced when the prefolding and postfolding components are the same polarity and little apparent polar wander has occurred between the times when the two magnetization components were acquired. To be resolved into a single characteristic magnetization, prefolding and postfolding components must have completely overlapping demagnetization spectra, making them inseparable by demagnetization. An overlap of both AF and blocking temperature spectra is most likely to occur by chemical addition of a secondary magnetic carrier that is identical to the primary phase [*McClelland-Brown*, 1982]. In some cases, demagnetization planes [e.g., *Halls*, 1976] may be used to separate components with imperfectly overlapping demagnetization spectra, but this analysis may require very detailed demagnetization, and it is successful only when one of the components is less dispersed than the other. Combinations of true synfolding components with prefolding components, with postfolding components, or with prefolding plus postfolding components can also produce a crossover geometry. The resultant vector of a combination of components of magnetization will not record a true geomagnetic field direction, and unless the different components can be resolved, it provides little direct geologic information.

Structural modification of rocks with prefolding magnetization (class 3, Figure 18) may also produce crossover geometries. Ignoring any direct effects of strain on the minerals carrying the magnetization, the crossover geometry

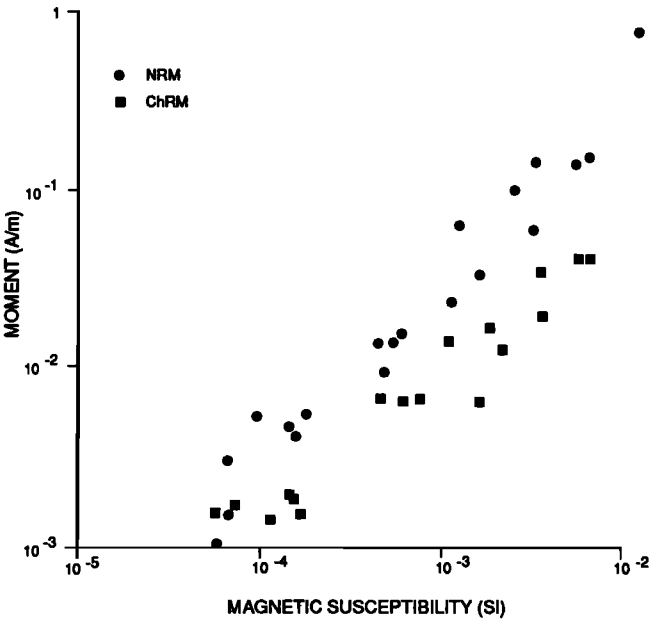


Fig. 17. Plot of site mean moment of natural remanent magnetization (NRM, circles) and characteristic magnetization (ChRM, squares) against magnetic susceptibility for sites of the Preuss Sandstone. Mean moment of characteristic magnetization and associated magnetic susceptibility are averaged from subsets of specimens subjected to thermal demagnetization for sites where the characteristic magnetization was well defined. The strong correlation between the parameters is expected when the detrital magnetite grains responsible for the magnetic susceptibility are also the predominant carriers of the NRM.

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Mechanisms to Produce Cross-over Geometry of Magnetization

Mechanism	Reference
Class 1. Acquisition of magnetization during folding	
CRM	McClelland-Brown, 1983 McCabe et al., 1983 Schwartz and Van der Voo, 1984 Kent and Opdyke, 1985 Miller and Kent, 1986a Benthien and Elmore, 1987 Bachtadse et al., 1987
TCRM	Granier et al., 1986
VpTRM	Kent, 1985
PRM	this study
Class 2. Combinations of different magnetization components	
Prefolding + postfolding	Schwartz and Van der Voo, 1984 Miller and Kent, 1988b
Prefolding + synfolding	this study
Prefolding + synfolding + postfolding	this study
Synfolding + postfolding	this study
Class 3. Structural modification of rocks with prefolding magnetization	
Indirect effect - oversteepened fold limbs	
1. Shear-slip folding	Tarling, 1983
2. Discontinuous strain (e.g., pressure solution)	Kodama, 1988
Direct effect - deflection of magnetization	
1. Magnetization as passive marker	Kligfield et al., 1983 Cogne' and Perroud, 1985 Hirt et al., 1986 Kodama, 1988
2. Rotation of magnetic minerals a. March model b. Layer-parallel shear & particle rotation	Cogne' et al., 1986 van der Pluijm, 1987 Kodama, 1988

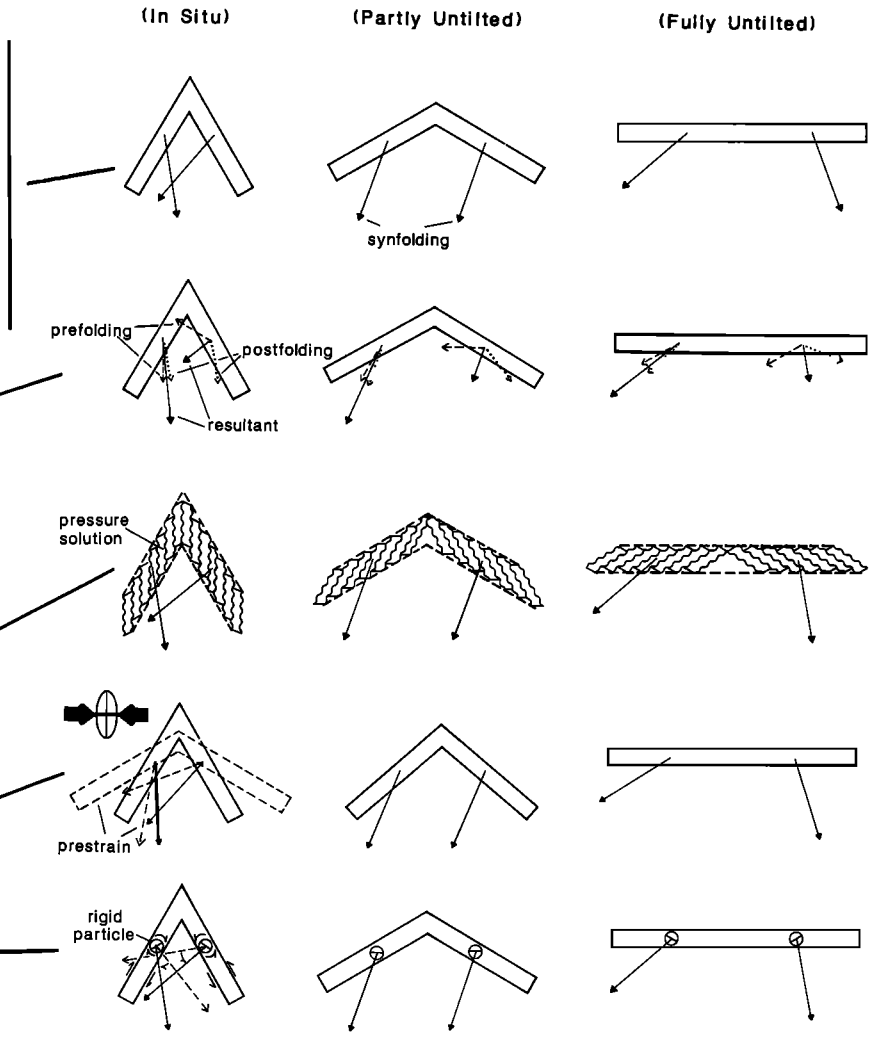


Fig. 18. Mechanisms that could produce a crossover geometry of magnetization, divided into three classes, for in situ, partly untilted, and fully untilted coordinates.

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TABLE 2. Analysis of Thermal Demagnetization Planes, Preuss Sandstone

	N	k_1	k_2	a_{min}	a_{max}	Oval Azimuth	Declination	Inclination
				Low-Temperature Planes				
In situ	110	−4.7	−1.2	6.5	11.1	99.2	358.3	69.6
Full correction	110	−2.9	−1.2	9.6	19.6	102.9	350.0	67.8
66% correction	110	−4.4	−1.0	6.9	10.8	104.9	358.6	67.3
				High-Temperature Planes				
Full correction	60	−5.0	−0.9	8.7	12.8	130.0	348.8	59.9
66% correction	60	−6.5	−0.9	7.3	10.0	127.0	339.0	64.6
				Combined				
Full correction	170	−3.4	−1.0	6.9	11.4	110.6	348.2	64.9
66% correction	170	−4.9	−0.8	5.3	7.3	107.5	348.2	66.5

N, number of specimens averaged; k_1 and k_2, Bingham concentration parameters; a_{min} and a_{max}, minor and major axes of 95% confidence oval; oval azimuth, azimuth of major axis of confidence oval; declination and inclination are of best intersection of demagnetization planes.

can be produced by overestimation of the amount of tilt correction required at a site when strain has oversteepened the dip of a fold limb beyond an angle due solely to rigid body rotation [*Tarling*, 1983; *Kodama*, 1988]. If the remanence is not directly affected, strain in this mechanism must be concentrated in discrete zones (e.g., spaced cleavage, pressure solution seams) thereby leaving most magnetic grains in the rock unaffected by the strain. Taking into account the effects of grain-scale strain on magnetization, some workers [*Kligfield* et al., 1983; *Facer*, 1983; *Cogne' and Perroud*, 1985; *Hirt et al.*, 1986; *Kodama*, 1988] have modeled a deflection of remanence directions during internal strain as that of passive markers (i.e., material lines in the rock). More recent models have considered the physical reorientation of the magnetic mineral grains during strain, assuming a model for the response of the grains to strain (e.g., March model), and have calculated the resulting deflection of the remanence [*Cogne' et al.*, 1986; *van der Pluijm*, 1987; *Kodama*, 1988]. All of these models can be used to predict that a prefolding remanence can be deflected into a crossover geometry, given realistic angular relations between strain, magnetization, and bedding during folding. In the unfolded but still-strained state the magnetization will not record the orientation of the geomagnetic field during its acquisition, although correct removal of strain effects may restore the magnetization to its prefolding orientation [*Kligfield et al.*, 1983; *Cogne' and Perroud*, 1985, 1987].

Preuss Magnetization: Is it synfolding?

Given the variety of mechanisms that can produce a crossover geometry, we must address whether the characteristic magnetization of the Preuss was acquired entirely or even partly during folding. Rock magnetic properties and diagenetic alterations [*Fishman et al.*, 1989] indicate that only detrital magnetite grains, not a mixture of detrital and secondary magnetite, carry the characteristic component in the Preuss. Thus evidence is lacking to support combinations of a postfolding or synfolding CRM component with a prefolding component (presumably depositional).

Selective remagnetization of some detrital magnetite grains may be considered to combine prefolding, synfolding, and (or) postfolding components. VpTRM acquisition could promote remagnetization of grains with short relaxation times, while grains with longer relaxation times retain their primary magnetization. In this case, however, theoretical studies suggest that two components should not display overlap of laboratory blocking temperature spectra unless the domain state or compositions in the magnetite grains vary widely [*McClelland-Brown*, 1982]; even under these conditions complete overlap is unlikely. It is prudent, nevertheless, to question whether there might be overlap of unblocking temperature spectra of multiple components in the Preuss. The angular discordance between characteristic components isolated by AF and thermal demagnetization at some sites in the Preuss (Figure 3) indicates that there is overlap of coercivity spectra of multiple components. For most sites, directions isolated by AF demagnetization are closer to a postfolding expected direction than those isolated by thermal demagnetization. This relation can be interpreted as the failure of AF demagnetization to effectively remove a postfolding VRM, given theoretical and experimental studies that show that thermal demagnetization is more efficient than AF demagnetization for removal of a VRM [*Dunlop*, 1973; *Dunlop and Stirling*, 1977]. A crossover geometry of magnetization in the Preuss for most sites could be produced if such a VRM component was also incompletely removed by thermal demagnetization.

To further test if incomplete separation of a postfolding or synfolding component from a prefolding component could produce the crossover geometry of magnetization in the Preuss, a demagnetization plane analysis was performed on the thermal demagnetization data. Demagnetization planes were fit to suitable low-temperature portions (typically 100°–400°C) of demagnetization paths in which the VRM was removed and the remanent magnetization moved toward the characteristic direction. In some samples a second demagnetization plane could also be fit to a high-temperature portion (typically 400°–620°C) of the demagnetization path in which the characteristic magnetization was removed and the remanent magnetization moved toward the direction of a hematite component. The dispersed nature of hematite components in the Preuss (Figure 7) allows a well-defined convergence of the high-temperature demagnetization planes on the characteristic component. To test for a prefolding versus synfolding age of the component obtained from this analysis, the best intersection of the demagnetization planes was examined for both fully corrected and partly (66%) corrected bedding tilts. For groups of both low-

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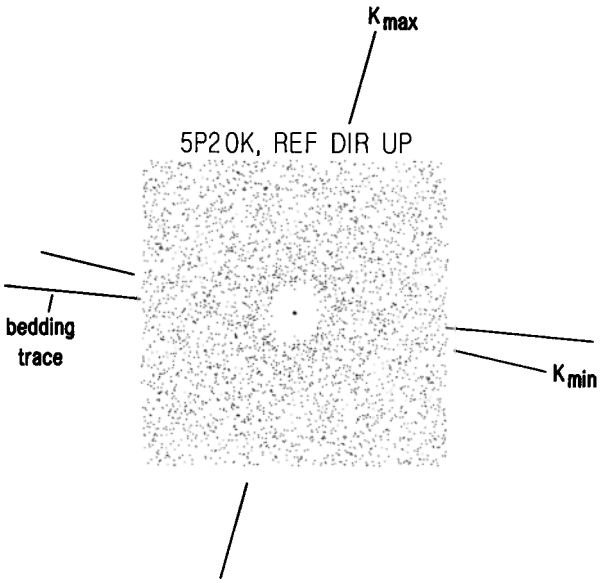


Fig. 19. Strain determination using center-center techniques [*Fry*, 1979] for a sandstone of the Preuss at site 20. The plane is perpendicular to the fold axis. The finite strain ratio is about 1.2 and has a maximum elongation oriented nearly perpendicular to bedding. Orientations of axes of the susceptibility ellipse on the plane are also shown for this sample.

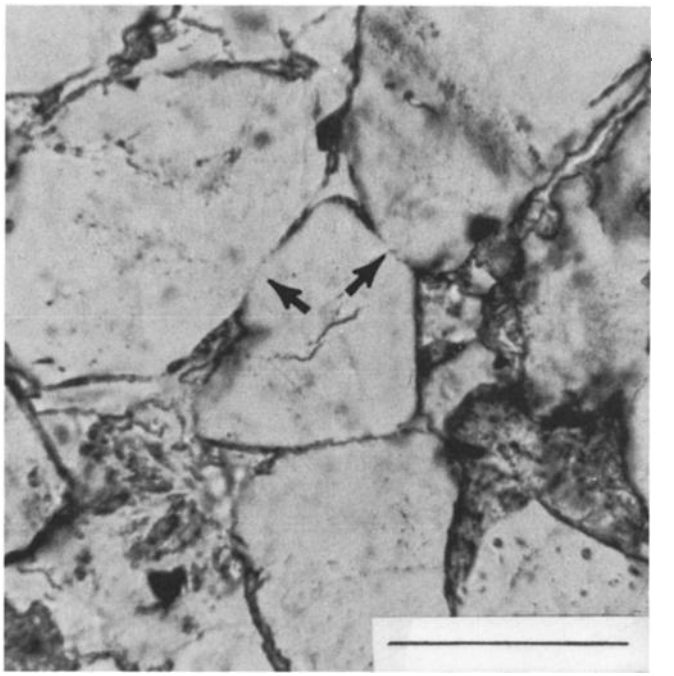


Fig. 20. Photomicrographs showing contacts between adjacent detrital quartz grains in the Preuss at site 12. Grain boundaries at the contacts lack coatings of diagenetic ferric oxide. The contacts are probably original depositional contacts that were shielded from deposition of diagenetic minerals in the early diagenetic history. Rigid body rotation did not occur between these grains during deformation. Bar is 100 μm. The section is vertical and perpendicular to the strike of bedding.

temperature and high-temperature planes, the best intersection is more tightly constrained for the 66% correction than for the full correction (Table 2). This supports a synfolding age for the isolated component. Moreover, the direction of the component isolated by both low-temperature and high-temperature planes is closer to Late Cretaceous and Tertiary expected directions than to middle Jurassic expected directions, regardless of the correction values (Table 2). This further suggests that no depositional remanence remains in the Preuss.

A simple structural modification of sandstones containing a prefolding remanence in the Preuss also cannot explain the crossover geometry of the magnetization. At the outcrop and hand-sample scale the sandstones show no discrete zones of concentrated strain (e.g., spaced cleavage, pressure solution seams). Moreover, petrographic observations suggest that internal strain of the sandstones was distributed throughout the rock at the sale of individual grains. Thus it is unlikely that strain of the sandstones caused a reorientation of bedding dips without directly affecting the magnetization of the Preuss. The AMS study suggests that most grain-scale strain of the sandstones occurred in response to layer-parallel compression that we interpret to have occurred in a subhorizontal compressive stress field while the rocks were still flat lying (i.e., prior to major folding). Grain-scale strain of the sandstones that accompanied later folding apparently was not sufficient to alter this earlier fabric. This is consistent with a flexural-slip mechanism, in which shear surfaces are restricted to layer boundaries, for folding of the sandstone layers in the Preuss. The flexural-slip style of folding is typical in competent units of layered sedimentary sequences of low-temperature portions of fold and thrust belts [*Faill*, 1973; *Suppe*, 1983; *Boyer*, 1986; *Mitra*, 1987]. This style is promoted by competency contrasts between sedimentary layers [*Ramsay and Huber*, 1987] such as those that probably occur between the sandstones, siltstones, and mudstones in the Preuss. A flexural-flow mechanism, in which bedding-parallel shear is distributed throughout the rocks, may occur in the less competent units. Flexural-flow folding is assumed for several models that predict deflection of prefolding magnetization into a crossover geometry during folding [*van der Pluijm*, 1986, 1987; *Kodama*, 1988].

Any deflection of remanent directions in response to prefolding layer-parallel shortening of the Preuss should have been of the same sense for samples from all of the sites. Such a deflection would not produce a crossover geometry of prefolding magnetization. Nevertheless, even if sandstones of the Preuss were strained during folding, it is unlikely that the magnitude of the strain would have been sufficient to noticeably deflect a prefolding remanence. AMS results show that magnetite grains in the Preuss have been affected by strain, but in no case has the magnitude of this strain completely masked the preexisting sedimentary fabric. Strain determinations using center-center techniques [*Fry*, 1979] from samples at several sites also indicate that finite strain in the sandstones is low. Strain ratios are typically only 1.1–1.2 (unstrained rocks have strain ratios of 1) for sections oriented perpendicular to fold axes (Figure 19). This amount of strain is far less than would be needed to account for the crossover geometry of magnetization directions. For untilted site mean directions that lie as far as 50° from expected Jurassic-Cretaceous directions, minimum strain ratios > 4.0 are required to produce this amount of deflection by flexural flow, assuming homogeneous strain [*van der Pluijm*, 1987].

A second model for deflection of remanence during flexural flow folding was also suggested by *van der Pluijm*

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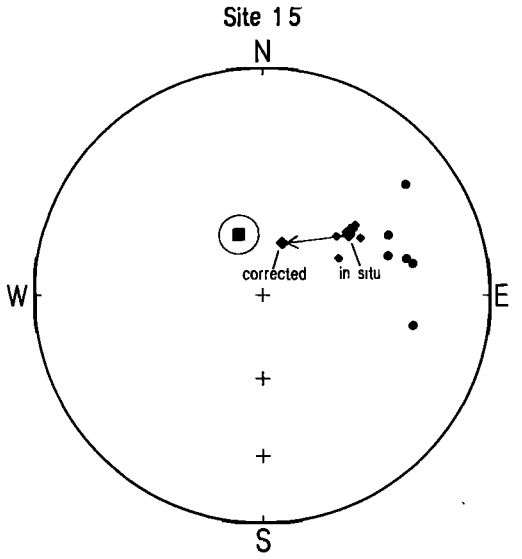


Fig. 21. Equal-area projection showing the in situ directions of characteristic magnetization isolated by thermal (diamonds) and AF (circles) demagnetization from samples at site 15. Also shown are the directions of the in situ and tilt-corrected mean of the directions (large diamonds) isolated by thermal demagnetization and the small circle path that the mean follows during untilting. The square is the 70-Ma expected direction for the thrust belt (approximate age of major movement on the Absaroka thrust) with its cone of confidence [from *Irving and Irving*, 1982]. Magnetizations from this site are atypical because the thermal mean does not move past the expected direction on untilting, and the magnetizations isolated by AF demagnetization are farther from the 70-Ma expected direction than those isolated by thermal demagnetization.

[1987], whereby particulate flow allows mineral grains distributed throughout layers on fold limbs to rotate in response to shear coupling between grains in bedding-parallel shear. Particulate flow [*Borradaile*, 1981], in which displacements occur at boundaries of individual rigid particles, may not be apparent in grain-scale strain measurements, and therefore its effects must be evaluated with caution. Intergrain sliding may have been important in the Preuss during earliest periods of layer-parallel shortening and may have promoted a loss of pore volume [*Fishman et al.*, 1989] as the grains were more tightly packed. However, further shortening led to intragranular strain of the framework grains and, along with abundant calcite cementation, this probably left the grains tightly interlocked and resistant to intergrain rotation during later folding. That intergrain rotation was not ubiquitous throughout the sandstones is also shown by the presence of points bare of diagenetic minerals between some framework grains (Figure 20). These features are interpreted as contact points between adjacent grains that were shielded from a coating of ferric oxide-clay minerals introduced during the earliest phases of the diagenetic history of the Preuss. That such contacts have remained intact illustrates that no rotation occurred between these grains. In summary, although the magnetite grains in sandstones of the Preuss have been affected by tectonic strain, neither the geometry or magnitude of this strain was sufficient to produce the observed crossover geometry of magnetization by deflection from a prefolding remanence direction.

We interpret the characteristic magnetization of the Preuss to have been acquired predominantly, if not entirely, during folding. That is, it is a true synfolding magnetization. The geometry of components of magnetization at site 15 (Figure 21) provides evidence that supports this conclusion.

Two atypical traits of the magnetization from site 15 allow an interpretation that tilting may have continued after acquisition of the characteristic magnetization. First, correction for a west dipping bedding tilt causes the site mean direction to move along a small circle toward the Late Cretaceous-early Tertiary expected direction. However, rather than crossing and moving past this direction, as is typical of most other site means (Figure 6), the site 15 mean direction is still far from the expected direction after complete correction for bedding tilt. Second, although characteristic components isolated by AF and thermal demagnetization are statistically distinct, as is common in other sites, the components isolated by AF demagnetization are farther from the Late Cretaceous-early Tertiary expected direction than those isolated by thermal demagnetization (Figure 21). This is opposite of the behavior observed at other sites. Both of these traits could be explained by a model in which the site was originally tilting eastward during acquisition of the characteristic magnetization, but was then later tilted back to the west. Continued tilting of the site after acquisition of the magnetization component supports a synfolding age for the component. Site 15 lies in the Absaroka thrust plate for which more than one discrete period of movement has been proposed [*Wiltschko and Dorr*, 1983]. Possibly, the magnetization at site 15 was acquired during an early period of thrust movement while the site was tilted eastward, but in subsequent movement the rocks were tilted westward. Dip reversals of bedding are predicted for certain portions of thrust plates in fault bend folding, so this model is structurally plausible. Nevertheless, we are not aware of specific thrust ramps at which the dip reversal could have occurred for site 15.

The evidence indicates that detrital magnetite grains of the Preuss acquired a secondary magnetization during folding. The mechanism(s) (i.e., CRM, TCRM, VpTRM) that caused remagnetization of these grains is not clear. Any mechanism that caused remagnetization of the Preuss should have accompanied deformation through space and time, because major folding is believed to have migrated across the thrust belt over several 10^7 years [*Armstrong and Oriel*, 1965; *Wiltschko and Dorr*, 1983]. Most paleomagnetic studies have explained synfolding magnetization by growth of new magnetic minerals and acquisition of CRM during folding [e.g., *McClelland-Brown*, 1983; *McCabe et al.*, 1983; *Miller and Kent*, 1986*a*]. Evidence for CRM acquisition by growth of magnetite in the Preuss is lacking. However, acquisition of a VpTRM during tectonic uplift would explain resetting of remanence in detrital grains of the Preuss. Furthermore, a sequential acquisition of the magnetization could be attributed to blocking of a VpTRM during local cooling of rocks in fault bend or fault propagation folds as individual thrust plates moved.

The major problem with using a VpTRM mechanism to explain the synfolding remanence of the Preuss is that the maximum paleotemperatures in the Preuss were not sufficient to completely reset any preexisting remanence carried by the detrital magnetite grains, based on current theoretical models of time-temperature relations for unblocking. Maximum paleotemperatures in the Preuss ranged from 150°C to as low as 75°C, based on values of vitrinite reflectance. In contrast, temperatures required by the theory of *Pullaiah et*

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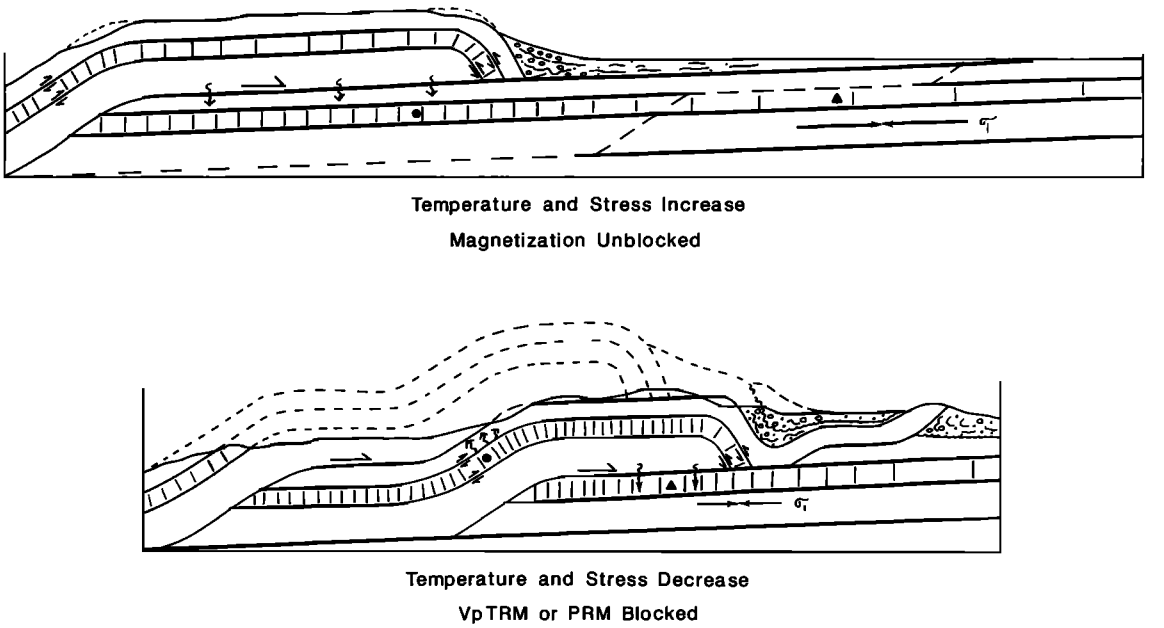


Fig. 22. Schematic model depicting the structural and thermal evolution of a foreland fold and thrust belt that may promote acquisition of secondary magnetizations during folding, as described in the text. Internal strain is depicted by near-vertical lines perpendicular to subhorizontal greatest principal stress σ_1. The first stage involves emplacement of a thrust plate over a given point (circle). Thrust emplacement promotes heat transfer (represented by wavy arrows) from the overlying thrust plate to the point and internal strain of footwall rocks. Movement of the next thrust plate up and over a thrust ramp causes cooling and folding of strata containing the point. Folding within the thrust plate proceeds by flexural slip (shown by small arrows), with little further internal strain. Cooling and decrease in the rate of internal strain during folding promote acquisition of a secondary magnetization in the thrust plate. With movement of the second thrust a second point (triangle) begins to undergo an increase in temperature and internal strain.

al. [1975] for complete resetting of magnetite with laboratory unblocking temperatures of 555°C are about 520°C if maintained for 10 Ma. The theoretical calculations of *Pullaiah et al.* [1975], however, probably overestimate the temperatures needed to reset magnetite [*Dunlop*, 1983; *Kent*, 1985; *Williams and Walton*, 1986]. Using more recent calculations [*Walton*, 1982; *Middleton and Schmidt*, 1982], temperatures predicted for complete resetting of the Preuss magnetite grains are about 470°C. Although neither of these theoretical analyses is strictly applicable to the magnetite grains in the Preuss because of their Ti content, it is unlikely that the presence of low amounts of Ti would have lowered the resetting temperatures to as low as 75°–150°C.

Stress-Enhanced Remagnetization

Although we concluded that the geometry and magnitude of internal strain of the Preuss was not sufficient to have deflected a prefolding magnetization into a crossover geometry by physical rotation of grains, we suggest that stress and (or) strain may have promoted remagnetization of the detrital magnetite grains in another way. Stress-enhanced acquisition of IRM has been termed piezoremanent magnetization (PRM [*Domen*, 1962]). Whereas PRM acquisition has been modeled theoretically and produced experimentally [e.g., *Domen*, 1962; *Ohnaka and Kinoshita*, 1968; *Nagata and Carleton*, 1968; *Kinoshita*, 1968; *Ohnaka*, 1969; *Nagata*, 1970], it has not commonly been recognized in paleomagnetic studies of natural rocks. This is not surprising because theoretical studies indicate that stress resets only the least stable portions of a preexisting magnetization [*Kern*, 1961], and therefore it may easily be mistaken for or overprinted by a VRM. Investigations of PRM have been carried out primarily to evaluate magnetization changes related to stress buildups in seismogenic zones [e.g., *Nagata*, 1970]. Experimental acquisition of PRM has been conducted by applying deviatoric stress, for short time periods, sufficient to produce only elastic strain. Under these test conditions the proportion of a preexisting thermoremanent magnetization reset by stress was small (typically <20%, [*Ohnaka and Kinoshita*, 1968; *Kinoshita*, 1968; *Ohnaka*, 1969]). However, little experimental evidence is available to assess the effects of long-term stress that might lead to permanent deformation of magnetite grains. Such conditions might more closely approximate the conditions that affected the Preuss. Experimental studies have shown that the magnitude of stress-induced changes of magnetic properties in titanomagnetites increases with increasing grain size and Ti content [*Nagata and Carleton*, 1968; *Kean et al.*, 1976]. Thus such effects might be favored in the multidomain magnetite grains of the Preuss.

We suggest two methods by which long-term stress and (or) strain may have promoted remagnetization of detrital magnetite grains in the Preuss: (1) stress-induced movement of crystal defects, and (2) stress-enhanced acquisition of VRM. Any intracrystalline plastic strain of the magnetite grains should occur by stress-induced movements of dislocations or point defects within the grains, analogous with deformation of other mineral types [*Hobbs et al.*, 1976]. Because the positions of domain walls in large multidomain grains are probably strongly influenced by the location of dislocations and point defects [*Stacey and Banerjee*, 1974], movement of these lattice defects during intracrystalline

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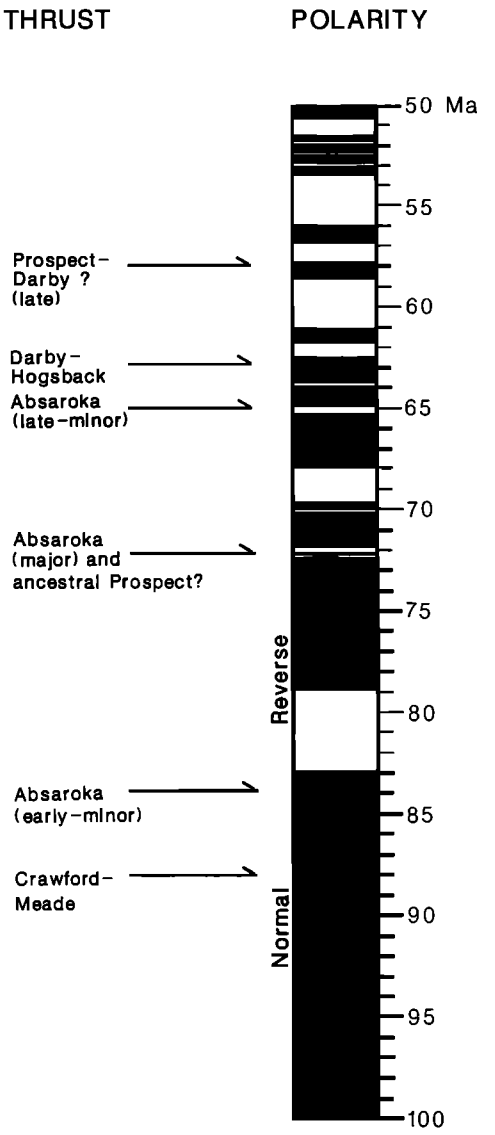


Fig. 23. Polarity time scale [from *Harland et al.*, 1982] during time span of deformation in the Wyoming-Idaho-Utah thrust belt. Age of movement on individual thrusts is estimated from *Wiltschko and Dorr*, [1983]. Possible ancestral movement on the Prospect thrust is related to westward thrusting of the Moxa Arch, as proposed by *Kraig et al.* [1987].

strain may also cause domain wall reorganization and remagnetization. This mechanism would be most analogous to PRM, although classical PRM occurs through domain wall movement as a direct result of stress rather than indirectly via intracrystalline strain. Unfortunately, whereas there is clear evidence of plastic intracrystalline deformation in other minerals in the Preuss (e.g., deformation bands in quartz, twinning in calcite), we have no independent evidence that the detrital magnetite grains have undergone plastic intracrystalline strain. Nevertheless, the cataclastic deformation of numerous homogeneous magnetite grains at several sites documents that the magnetite grains in the Preuss experienced high levels of internal stress.

Stress-enhanced VRM acquisition, similar to that suggested by *Stott and Stacey* [1960], could also be responsible for Preuss remagnetization. From a thermodynamic standpoint, maximum paleotemperatures in the Preuss should not alone have sufficiently raised the free energy base level of magnetite grains to allow thermal fluctuations to jump the potential energy barriers required for domain wall movement, the mechanism for VRM acquisition for multidomain grains. However, deviatoric stress is also known to raise the internal free energy of mineral grains [*Hobbs et al.*, 1976]. We suggest that the effects of temperature and stress, when combined, may raise the free energy in magnetite grains sufficiently to overcome the potential energy barriers to VRM acquisition. This would allow VRM acquisition at temperatures lower than predicted by theoretical calculations that do not account for stress [*Pullaiah et al.*, 1975; *Walton*, 1982]. Experiments of VRM acquisition in samples under stress should be conducted to test this hypothesis. In addition, experimental work [*Boyd and Fuller*, 1982] suggests that stress may cause multidomain magnetites to nucleate new domain walls. An increase in domains could lead to transdomain VRM acquisition [*Moon and Merrill*, 1986] if increasing the number of domains reduces the overall energy state for the grains.

Model for Synfolding Acquisition of Preuss Magnetization

We propose that acquisition of synfolding magnetization in the Preuss Sandstone was caused by changes in the structural and thermal regimes of beds during deformation (Figure 22). As a more westerly thrust plate overrode the Preuss at any particular point, that point was buried to its greatest depth and, with thermal reequilibration [*Angevine and Turcotte*, 1983], it soon reached its maximum temperature. Increased confining pressure allowed an increase in the deviatoric stress that built up before brittle failure occurred [*Sibson*, 1983]. The increased deviatoric stress and elevated temperature promoted internal strain of the Preuss in response to horizontal, layer-parallel compression. Elevated temperature, increased deviatoric stress, and internal strain promoted unblocking of primary magnetization in the detrital magnetite grains. Later, as the point was incorporated in the next thrust plate that ramped toward the surface, it was uplifted and incorporated in major fault propagation or fault bend folds. Uplift induced cooling of the point. Structural styles changed as the rocks began to fold. Internal strain within the rocks diminished as further shortening was accommodated predominantly by flexural-slip folding, with bedding-parallel shear confined to bedding planes. The combination of cooling and decrease of internal strain during folding both acted to block in secondary magnetization that, until that time, had been able to reorient to any relative changes of the geomagnetic field direction. With further tightening of these folds the secondary magnetization acquired a crossover geometry. This process was repeated on each thrust plate and thus allowed acquisition of the synfolding magnetization to accompany thrusting through space and time in the thrust belt.

Remaining Problems

An important question that remains poorly answered in our explanation of synfolding magnetization in the Preuss is the lack of reverse-polarity magnetization in the Preuss. The ages estimated for movements on thrusts [*Wiltschko and Dorr*, 1983] span times of mixed geomagnetic polarity (Fig-

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ure 23), [*Harland et al.*, 1982]. If the characteristic magnetization of specimens in the Preuss recorded discrete, instantaneous field directions during folding, some reverse-polarity field directions would be expected. This is true for any mechanism invoked to record synfolding magnetization (class 1, Figure 18). However, other mechanisms that might have produced the crossover geometry of magnetization (classes 2 and 3, Figure 18) also cannot account for the lack of reverse-polarity magnetization, given that a prefolding depositional remanence in the Preuss would be expected to have dual polarities like other Middle Jurassic rocks [*Ogg et al.*, 1984; *Steiner et al.*, 1985, 1987]. Combination of prefolding and other components (class 2) would not produce a crossover geometry if the prefolding component had dual polarities. Structural modification of rocks (class 3) would not change dual polarities of a prefolding magnetization.

We do not have a documented explanation for the lack of reverse-polarity magnetization in the Preuss. One possible explanation, however, involves acquisition of the characteristic magnetization over an extended rather than a discrete time span during folding. If the characteristic magnetization was acquired over several polarity intervals, only the predominant polarity during remanence acquisition might be apparent during demagnetization. Prolonged acquisition might be viable for stress-enhanced remagnetization because of inhomogeneous stress levels experienced among detrital magnetite grains in the Preuss. An inhomogeneity of stress is suggested by observations in some samples that magnetite grains caught between contacts of adjacent grains are highly fractured (Figure 8*f*), whereas nearby grains of similar composition escaped noticeable brittle fracture. If stress levels in individual grains varied widely, stress-enhanced remagnetization of all grains may have been prolonged. This interpretation can explain the normal polarity of site means from the Crawford-Meade and Absaroka thrust plates which moved during Late Cretaceous periods of predominantly normal polarity (Figure 23). However, latest movements for the eastern plates occurred during Paleocene to Eocene periods of predominantly reversed polarity. We have to invoke remagnetization of the eastern sites during the proposed Late Cretaceous ancestral movement of the Prospect thrust [*Kraig et al.*, 1987] to explain their normal polarity using this interpretation.

CONCLUSIONS

The Preuss Sandstone of the Wyoming-Idaho-Utah thrust belt contains characteristic magnetization whose directions group most tightly at an intermediate stage of unfolding. Petrographic analysis and rock magnetic properties indicate that only detrital titaniferous magnetite grains carry this magnetization. Remagnetization of these grains probably occurred at low temperatures (<150°C) during folding. These results, along with petrographic and AMS evidence that the detrital magnetite grains have been affected by stress and strain, suggest that the low-temperature remagnetization was tectonically enhanced by stress-accelerated acquisition of a VRM and (or) by stress-induced intracrystalline strain and accompanying domain wall movement (PRM). The locus of acquisition of synfolding magnetization in the Preuss migrated with deformation in the thrust belt. We suggest that synfolding magnetization was blocked as individual thrust plates cooled and folded as they ramped upward. A lack of reverse-polarity magnetization in the Preuss is tentatively explained as the record of the predominant polarity during Late Cretaceous and early Tertiary (?) phases of deformation.

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